Precambrian Research 268 (2015) 194–211

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

A revised paleomagnetic pole from the mid-Neoproterozoic Liantuo Formation in the Yangtze block and its paleogeographic implications

Xian-qing Jing[a], Zhenyu Yang[b,c,*], Yabo Tong[b,c], Zhirui Han[b]

[a] School of Earth Sciences and Engineering, Nanjing University, Nanjing 210093, China
[b] Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Land and Resource, Beijing 100081, China
[c] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China

ARTICLE INFO

Article history:
Received 10 November 2014
Received in revised form 5 July 2015
Accepted 18 July 2015
Available online 28 July 2015

Keywords:
Paleomagnetism
Neoproterozoic
Liantuo Formation
South China block
Australia
Paleogeographic reconstruction

ABSTRACT

The paleogeographic relationship between South China and Australia during the Neoproterozoic is still hotly debated. Although a series of propositions for the close proximity between South China and Australia are suggested, their relative positions are varied. To better constrain the paleoposition of South China, we have carried out a new paleomagnetic study from the Neoproterozoic Liantuo Formation in two localities, Yichang and Changyang, in Hubei province, China. Stepwise thermal demagnetization reveals that a late Cretaceous remagnetization (component 'L', 350–500 °C) was removed from most of the samples. A medium component ('M') isolated mainly between 350 °C and 600 °C, but in some cases with an upper limit of 640 °C, yields a mean direction of Dg = 59.5°, Ig = 64.6°, α95 = 3.7° in situ; and Ds = 81.6°, Is = 61.5°, α95 = 4.1° after tilt correction. A steeper medium-high temperature component, 'B', was separated from 71 specimens with almost the same temperature range of 'M' (Dg = 341.3°, Ig = 78°, α95 = 5.6° in situ; and Ds = 78.3°, Is = 83.8°, α95 = 6.9° after tilt correction) and is similar to the 'B' component obtained by Evans et al. (2000). In addition, a high temperature component 'H' (Ds = 102.5°, Is = 62.4°, α95 = 3.7° after tilt correction) with both reversed and normal polarity was revealed up to 690 °C, yielding a pole at 12.7° N, 157.4° E, with dp/dm = 4.5°/5.8°. A change of magnetic polarity, with 5 magnetozones defined, is revealed in our sampling section, in which a reversed-normal polarity succession in the upper part of the section may be correlative with that obtained in a section 6.5 km away by Evans et al. (2000), demonstrating the primary nature of the 'H' component. Combining our results with those of Evans et al. (2000) yields a grand mean pole from the upper Liantuo Formation at 13.2° N and 155.2° E with A_{95} = 5.3°. Considering the distribution of our sampling sites, and together with the results of a detailed recent SIMS U–Pb zircon study on the Liantuo Formation in the Three Gorges area (Lan et al., 2015), an age of 720 Ma was assigned to component 'H' and its corresponding pole. After carefully reevaluating the Neoproterozoic apparent polar wander paths (APWP) from Australia and South China, we suggest that the South China block was located to the northwest of Northern Australia during the Ediacaran period. Subsequently, the South China block moved to the northwest of Western Australia through counter-clockwise rotation of the Northern Australian craton at the later time of 550 Ma, or in the middle Cambrian.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

Estimation of the paleoposition of south China in the late Proterozoic is potentially important for reconstructing the Rodinia supercontinent (Li et al., 1995). Various disciplines and methods have been used to constrain its paleo-positions, including geochronology, geochemistry, sedimentary basin analysis, analysis of detrital zircon provenance, petrology, paleontology, tectono-stratigraphy and paleomagnetism. The application of these various approaches has resulted in the production of several reconstruction models during the past two decades. In these reconstructions, the South China block (SCB) either moved quickly from the central part of Rodinia in 750–780 Ma (the 'missing-link' model, Li et al., 1995, 2008; Li and Evans, 2011), and then to the north of Australia (Li and Evans, 2011) or northeast of Australia by 650 Ma (Li et al., 2008), or to the northwestern part of Australia during post-635 Ma and

* Corresponding author at: College of Resources, Environment and Tourism, Capital Normal University, 105 Xi San Huan Bei Road, Haidian District, Beijing 100048, China. Tel.: +86 10 88815152; fax: +86 1068422326.
E-mail address: yangzhenyu@cags.ac.cn (Z. Yang).

http://dx.doi.org/10.1016/j.precamres.2015.07.007
0301-9268/© 2015 Elsevier B.V. All rights reserved.

early Paleozoic time (Yang et al., 2004; Zhang et al., 2013, 2015). On the other hand, a relatively stationary relationship between the SCB and Australia has also been suggested based on late Proterozoic, middle Cambrian and Silurian paleopoles of south China (Zhang and Piper, 1997; Yang et al., 2004), in which south China was situated in the northwestern part of Australia between the late Proterozoic and early Paleozoic.

It is clear that the paleogeographic reconstruction of South China during the late Proterozoic and early Paleozoic is hampered by the lack of reliable paleomagnetic poles. Although paleomagnetic studies of the Neoproterozoic and early Paleozoic rocks of South China have been carried out for more than three decades (Lin et al., 1985a,b; Zhang et al., 1983), only a few high quality poles are available (Evans et al., 2000; Huang et al., 2000; Yang et al., 2004; Zhang et al., 2013). A high quality 748 Ma 'Z1′ pole reported by Evans et al. (2000) was used by Yang et al. (2004) in the reconstruction of South China and Australia. Although in their reconstruction the middle Cambrian and Silurian poles are well-matched from both the Yangtze and Australia blocks, the 'Z1′ pole is not consistent with the 755 Ma Australian MDS pole. Based on the lower Triassic paleomagnetic results for the Zigui area of Shen et al. (1999), Yang et al. (2004) proposed that the discrepancy could be caused by the clockwise rotation of the sampling area at Yichang (~20°). However, Zhang et al. (2013) listed several lines of evidence indicating that no rotation occurred in the Z1 sampling region.

Recently, Zhang et al. (2013) reported a new Nantuo pole (635 Ma) that is not significantly different to the Liantuo Z1 pole, which implies that the Z1 pole may be much younger than previously thought. In addition, the age discrepancy between the Z1 pole and MDS pole must be considered in the course of directly comparing the rotated Z1 pole with the MDS pole within the Australian framework. The Liantuo Formation was first dated using SHRIMP U–Pb dating of twenty-three zircons from an interbedded tuff with a discordia lower intercept age of 748 ± 12 Ma (Ma et al., 1984). However, recent studies report generally younger ages from the Liantuo Formation (Lan et al., 2015; Zhang et al., 2006; Cui et al., 2014; Liu et al., 2008; Hofmann et al., 2011).

In order to verify whether or not there was local rotation of the Yichang area, and to determine the long-term directional change within the Liantuo Formation from the bottom to the top, we carried out a systematic paleomagnetic study of two separate sections in the Tianjiayuan and Wangjiapeng areas. The former is located 6.5 km from the section studied by Evans et al. (2000).

2. Geological setting and paleomagnetic sampling

The study areas are located in Yichang and Changyang, Hubei Province, in the northern part of the Yangtze Block, where typical Precambrian sedimentary sequences have been mapped by geological survey (Fig. 1). The stratigraphy of the study areas includes the Liantuo Formation, the Nantuo Formation, the Doushantuo Formation and the Dengying Formation, from older to younger. The Liantuo Formation consists of thick-bedded, red purple conglomerate feldspathic sandstone and quartz sandstone in the lower part; and clastic sandstone, feldspathic sandstone, silty sandstone and crystal-vitric tuff in the upper part. The Nantuo Formation (654 Ma–635 Ma, Zhang et al., 2008) consists predominantly of dark green and minor red to purple tillites that disconformably overlie the Liantuo Formation. The disconformity between the Liantuo and the Nantuo Formations may represent a ~60 Ma deposition gap (ca. 710–650 Ma). The Doushantuo Formation overlies the Nantuo Formation with an insignificant disconformity, and comprises dolomite with interlayered phosphatic shale, which was dated to ca. 635 Ma–551 Ma (Zhang et al., 2005; Condon et al., 2005; Liu et al., 2008). The Dengying Formation, continuously developed over the Doushantuo Formation, consists mainly of thick-bedded dolomite and limestone.

A total of 480 samples were collected from 40 sites using a gasoline-powered drill, including 412 (33 sites) comprising red and a few green siltstone-sandstone samples from the Tianjiayuanzi area (TL) at Yichang (GPS: 111.069° E, 30.819° N) and 68 (7 sites) comprising fine-medium-grained red-colored samples from the Wangjiapeng section (WL) at Changyang (GPS: 111.069° E, 30.538° N).

The TL area includes three sub-sections at Huajipo, Toudingshi and Tianjiayuanzi (Fig. 2). We collected samples from 9 sites from the bottom to top of the Liantuo Formation at the Huajipo sub-section (Fig. 2). Because of heavy vegetation cover in the lower part of the Liantuo Formation, the samples were collected only from the uppermost parts of two other sub-sections, where the sampling layer positions were determined from the distances between the site and the boundary between the Liantuo-Nantuo Formations (Fig. 2). The Tianjiayuanzi sub-section was located on the south bank of the Yangtze River, close to the sampling locality studied by Evans et al. (2000) (Fig. 1).

The dip attitudes of the Liantuo formation are mostly directed to the SE quadrant with dip angles less than 30° in the Tianjiayuanzi sub-section (Fig. 1). Although the dip attitudes of the strata in the Wangjiapeng section are variable, the trend is approximately toward the NW and NNW (Fig. 1).

3. Laboratory techniques

In order to characterize the magnetic mineralogy, representative specimens were chosen for various rock magnetic experiments. Acquisition of isothermal remanent magnetization (IRM) was imparted using an ASC IM-10-30 impulse magnetizer and measured with a JR-6A spinner magnetometer. To further constrain the unblocking temperatures of the magnetic minerals, representative samples were magnetized sequentially along *Z*-, *Y*-, and *X*-axes at fields of 2.4 T, 0.4–0.5 T, and 0.12 T, respectively, and were then subjected to stepwise thermal demagnetization (Lowrie, 1990). The anisotropy of magnetic susceptibility (AMS) was measured using the KLY-3S Kappa-bridge.

Stepwise thermal demagnetization of pilot samples was performed using an ASC-TD48 oven in order to isolate the characteristic remanent magnetization (ChRM). After analyzing the behavior of the pilot samples, progressive thermal demagnetization was carried using 14–18 steps at a 50 °C interval for low temperatures (<300 °C), and a 20 or 30 °C interval for high temperatures (>300 °C) up to 600–660–670 °C, and finally with a 10 or 15 °C interval to 685–690 °C. The remanence of the specimens was measured using a 2G Enterprises Inc. cryogenic rock magnetometer (2G-755) housed in a magnetically shielded room. The magnetic remanence directions were analyzed using the principal component analysis method (Kirschvink, 1980) and the site-mean directions were calculated using Fisher spherical statistics (Fisher, 1953) or using the intersection of remagnetization great-circle method (McFadden and McElhinny, 1988). Palaeomagnetic software packages PMGSC (by R. Enkin) and PaleoMac (Cogne, 2003) were used to perform data analyses and produce related figures. All of the magnetic measurements were made at the Palaeomagnetism Laboratory of Nanjing University.

4. Paleomagnetic results

For most of the samples, the IRM acquisition curves show a relatively rapid increase of intensity up to 800–900 mT, but with saturation not being attained until about 1.5–1.8 T (Fig. 3C, D, G, H). This suggests that the samples contain predominantly

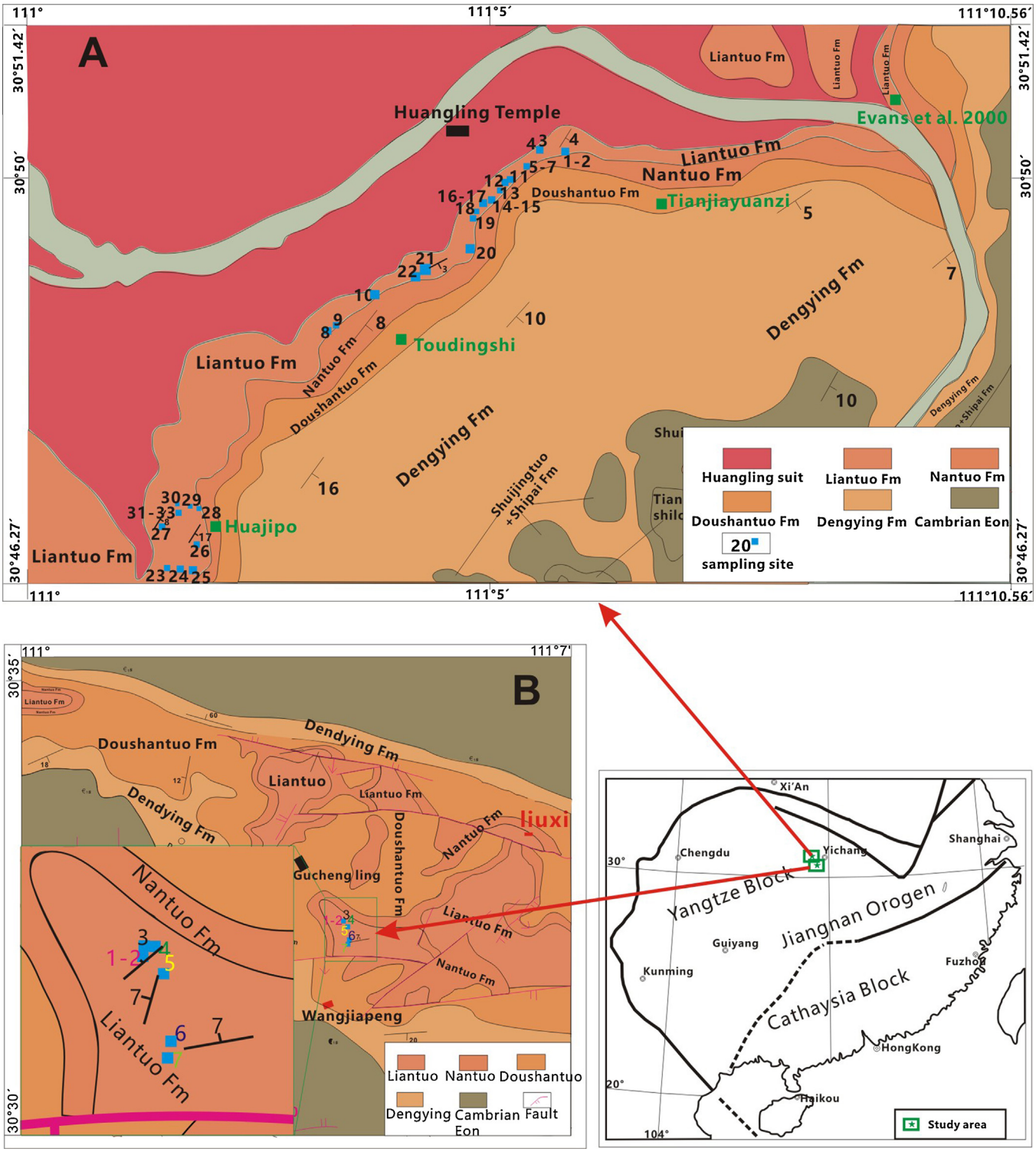


Fig. 1. Tectonic framework of South China, and geological map of the sampling areas at Zigui (A) and Changyang (B), Hubei Province. The sampling site of Evans et al. (2000) is also shown for comparison.

intermediate to high-coercivity magnetic minerals and minor low-coercivity magnetic phases. Stepwise thermal demagnetization of IRMs reveals that high- and intermediate-coercivity components are dominant, with the magnetization decreasing sharply at ~680 °C (Fig. 3A, B, E, F), indicating the presence of hematite. A significant low coercivity component decays rapidly around 580 °C (Fig. 3A and E), suggesting the presence of magnetite (Lowrie, 1990).

Based on their significantly different demagnetization characteristics, the samples from the two study areas are treated separately.

4.1. Tianjiayuanzi area (TL) (111.069° E, 30.819° N)

Four magnetic components, designated 'L', 'M', 'B' and 'H', are evident in the stepwise thermal demagnetization results from

the samples from the TL section (Figs. 4–6). Most of the samples are characterized by three components; however, several samples exhibit a single 'H' component or a composite of two components ('L-H' and/or 'L-M-erratic').

Component 'L' is separated below 350–500 °C (Figs. 4–6) at 31 sampling sites. The mean direction (Dg = 13.8°, Ig = 54.0°, α_{95} = 3.7°, N = 31 sites, before tilt correction) is close to those previously determined for Cretaceous poles from south China (Fig. 7, Table 1) (Yang and Besse, 2001). This direction is also consistent with the widespread overprinting component revealed in Paleozoic sedimentary rocks in the Three Gorges region, indicating a Cretaceous remagnetization event related to the late phases of the Yanshanian orogeny (Kent et al., 1987).

Component 'M' is determined mostly between 350 °C and 600 °C (Figs. 5 and 6, in some cases the upper limit is 640 °C) in samples from 25 sampling sites in the middle and upper parts of the TL section (Fig. 8). The mean direction is Dg = 59.5°, Ig = 64.6°, α_{95} = 3.7° in situ; and Ds = 81.6°, Is = 61.5°, α_{95} = 4.1° after tilt correction (Fig. 8, Table 1), and their corresponding poles are significantly different to the poles for the SCB since the Cambrian (Fig. 7).

We also separated a less common middle-high temperature component 'B' (similar to component B obtained by Evans et al. (2000)) in 71 specimens from 6 sites of the TL section. In general, the component 'B' and component 'M' are defined almost within the same temperature range (Figs. 4–6). However, several specimens remain unblocked up to 670 °C (Fig. 6). While components 'B' and 'M' coexist, component 'B' always has a higher unblocking temperature (Fig. 5). The mean direction of component 'B' is slightly steeper than that of Evans et al. (2000) (Dg = 341.3°, Ig = 78°, kg = 145.3, α_{95} = 5.6° in situ; and Ds = 78.3°, Is = 83.8°, ks = 94, α_{95} = 6.9° after tilt correction, kg/ks = 1.546) (Fig. 8, Table 1). The corresponding pole, either before or after bedding corrections, does not resemble the Phanerozoic paleopoles of the South China block (Fig. 7).

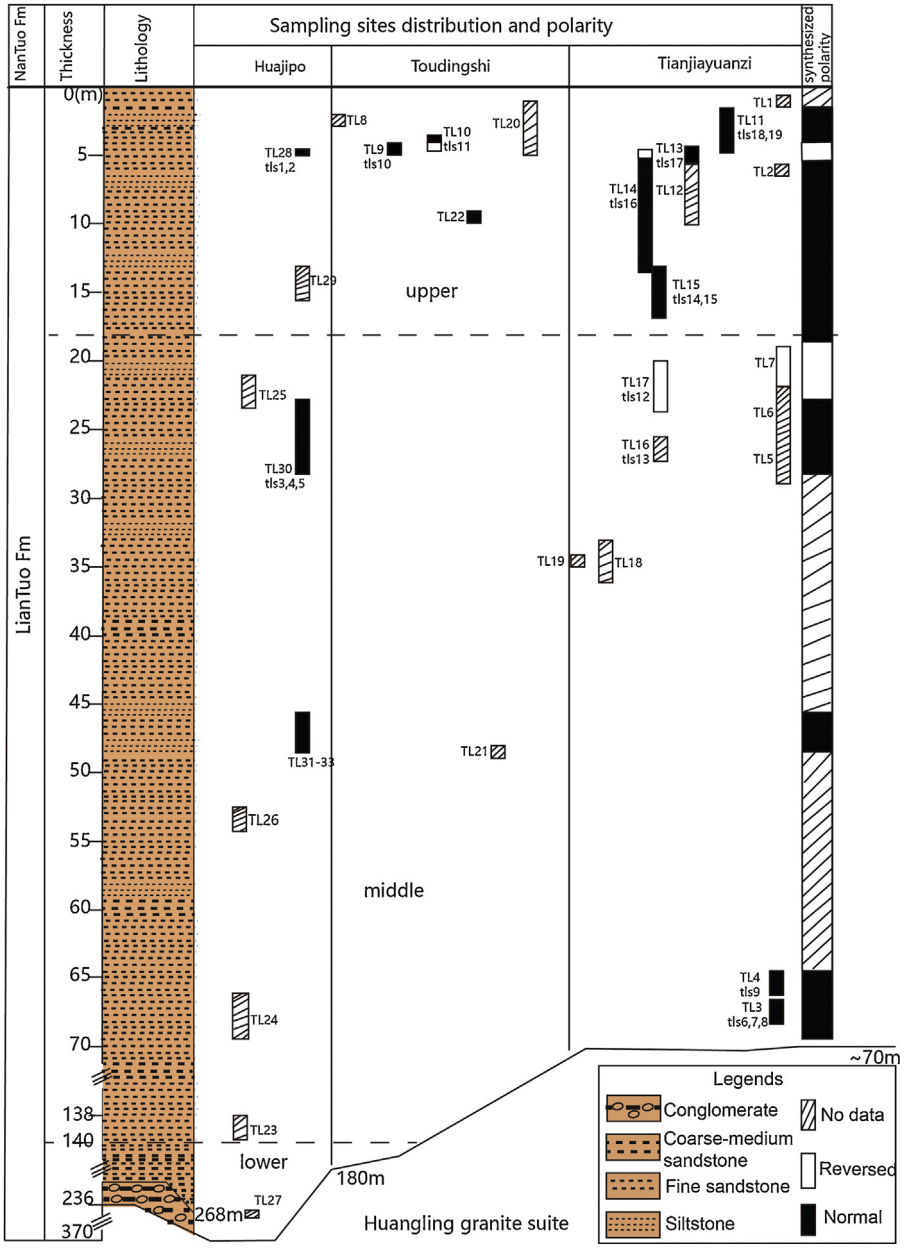


Fig. 2. Stratigraphic column of the Liantuo Formation and detailed sampling layer positions of three sub-sections at Yichang. The composite magnetic polarity sequence is also shown.

Table 1
Site-mean directions calculated from the palaeomagnetic results of different temperature components of samples from the Liantuo Formation at Yichang, South China (mean GPS: 111.069° E, 30.819° N).

Site	Comp.	n	Dg	Ig	Ds	Is	k	α95
TL9	H	5	47	82.2	122.6	71.8	135.8	6.6
TL10	H	6	255.8	−66	280.4	−58.1	28.3	12.8
TL11	H	4	86.4	68.8	111.1	60.6	60.4	11.9
TL13	H	12	96.6	66.7	110	58.1	59.5	5.7
TL14	H	11	275.1	−65.9	286.3	−59	127.6	4.1
TL15	H	14	67	67.6	90.3	62.3	73	4.9
TL22	H	10	87.8	65.1	93.4	63.6	52.6	6.7
TL28	H	7	58.1	65.6	74.2	63.2	31	11
Mean H component of upper part		8	79	69.1	100.6	62.7	103.1	5.5
Paleomagnetic pole:				14.1 N, 157.8 E, 6.7/8.6, A95 = 7.6 (*Tilt corrected*)				
TL3	H	8	95	66	105.1	64.1	248.5	3.5
TL4	H	12	87.5	69.9	117.7	60.8	59.1	5.7
TL7	Hg	4	253.1	−71.6	271	−67.2	344.5	9.5
TL17	Hg	9	285.9	−60.7	291	−50.2	13.5	15.3
TL30	H	8	76.8	69.5	99.9	62	102.5	5.5
TL33	H	5	76.4	73.2	99.6	66.1	20.6	17.3
Mean H component of middle part		6	87.7	68.9	105	62	116.1	6.2
Paleomagnetic pole:				10.9N, 156.8E, 7.5/9.6, A95=8.5 (*Tilt corrected*)				
Mean H components (Upper + middle parts of Liantuo Formation)								
		14	82.8	69.1			122.1	3.6
					102.5	62.4	115.4	3.7
Paleomagnetic pole:				28.2 N, 154.2 E, dp/dm = 5.2/6.1 (*In situ*)				
				12.7 N, 157.4 E, dp/dm = 4.5/5.8 (*Tilt corrected*)				
Mean L components								
		31	13.8	54	28.9	60.1	39.5	3.7
Paleomagnetic pole:				77.8 N, 179.7 E, dp/dm = 4.1/5.9 (*In situ*)				
TL1	M	15	47	58	53	56.8	14.6	10.4
TL3	M	10	78.9	53	85.5	52.6	48.9	7
TL4	M	3	59.7	66	112.3	59.9	48.9	17.8
TL6	M	9	55.7	60.9	76.9	57.4	30.2	9.5
TL7	M	11	64.6	59.7	76.7	56.7	21.3	10.1
TL8	M	5	54.8	57	71.5	66.6	34.8	13.1
TL9	M	5	34.5	78	107.6	72.9	89	8.2
TL10	M	9	50.5	69.2	86.9	66	35	88
TL11	M	9	54.3	71.4	93.4	68.9	97.7	5.2
TL12	M	7	59.5	60.9	74.5	57.9	26.8	11.9
TL13	M	11	45.8	68.9	76.6	68	27	9
TL14	M	5	61.4	70.7	83.5	67.3	94.3	7.9
TL15	M	16	59.5	68.4	85.6	64.2	34.9	6.3
TL16	M	13	72.8	67.4	103.1	62.1	35.2	7.1
TL17	M	4	52.6	78	87.3	71.7	431.7	4.4
TL18	M	10	53.2	61.6	76.1	59	20.3	11
TL19	M	4	64	68.5	91.2	61.8	55.6	12.4
TL21	M	8	84.5	38.8	89.9	29.7	10.7	17.8
TL22	M	7	57.6	77.9	71.5	77.7	116.4	5.6
TL23	M	10	56.1	60.1	69.9	53.8	43.9	7.4
TL28	M	9	51.2	64.3	67.2	62.9	52.2	7.2
TL30	M	11	64	67.3	89	63	34	7.9
TL33	M	9	64.7	64.7	83.9	59.9	20.8	11.6
TL31	M	20	32.7	56.6	57.5	54.1	25.2	6.6
TL32	M	11	69.3	53.7	82.5	48.5	24.4	9.4
Mean M components								
		25	59.5	64.6			61.4	3.7
					81.6	61.5	49.9	4.1
Paleomagnetic pole:				42.2 N, 164.3 E, dp/dm = 4.8/5.9 (*In situ*)				
				26.1 N, 165.2 E, dp/dm = 4.9/6.3 (*Tilt corrected*)				
9	B	11	332	82.8	145.3	79.2	86.1	4.9
10	B	12	317.3	72.4	275.7	86.3	39.5	7
14	B	6	331.7	83.9	120.7	86.8	103	6.6
17	B	13	13.9	76.8	62.4	76.7	107.5	4
26	B	19	336.1	74	55.1	80.7	61.7	4.3
31	B	10	354.4	73.6	62.2	76.7	125.2	4.3
Mean B components								
		6	341.3	78			145.3	5.6
					78.3	83.8	94	6.9
Paleomagnetic pole:				52.2 N, 99.3 E, dp/dm = 9.9/10.5 (*In situ*)				
				32.5 N, 125.3 E, dp/dm = 13.3/13.6 (*Tilt corrected*)				

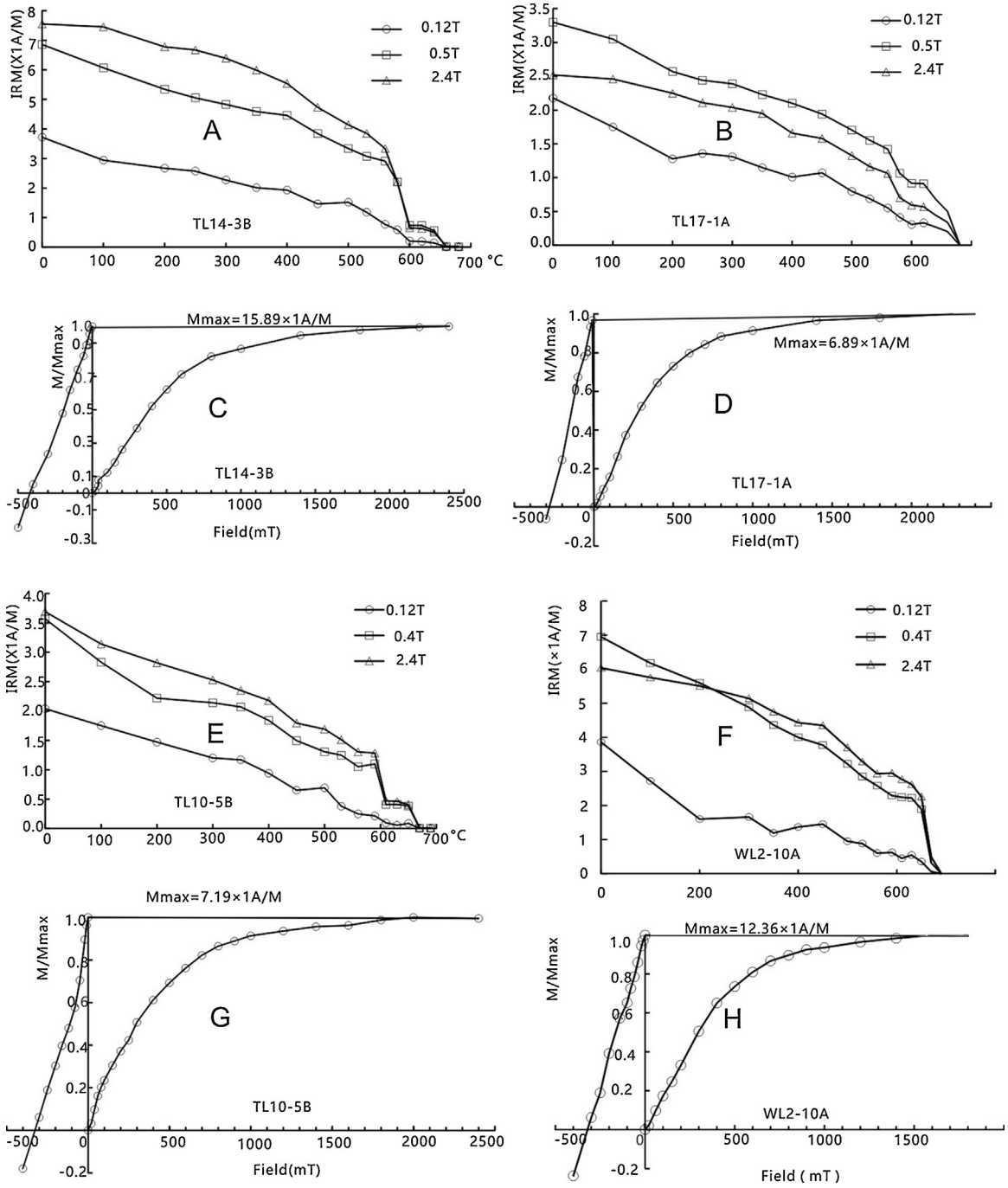


Fig. 3. Normalized isothermal remanent magnetization (IRM) acquisition curves and backfield demagnetization (C, D, G, H). Three-component IRM stepwise thermal demagnetizations (A, B, E, F) of representative samples showing unblocking temperatures around 680 °C. Orthogonal IRMs were imparted in different fields: 0.12 T (circles), 0.4 T (squares), and 2.4 T (triangles), respectively.

A high temperature component 'H', with either reversed or normal polarity (Figs. 4–6), is revealed up to 690 °C from these red bed samples in a total of 14 sites (115 specimens). The presence of high-coercivity hematite is evidenced by the IRM acquisitions, thermal demagnetization and thermo-remanent analyses (Fig. 3). Component 'H', decaying linearly to the origin, could be determined at 12 sites (102 specimens). However, the demagnetization directions become erratic after 620–640 °C in a few samples (sites 7 and 17) (Fig. 4), which prevents the determination of the final directions. In this case, several samples exhibited well-defined great circle paths at temperatures below 640 °C which are fitted and shown on the stereographic plot (Fig. 4).

The site-mean direction was then calculated using the intersection of sector-constrained great circles, combined with directly observed directions, using the statistical method developed by McFadden and McElhinny (1988) (Fig. 4). The directions of component 'H', with an East-East-South declination (downward inclination) and a nearly antipodal West-West-North declination (upward inclination) after tilt correction are revealed at 14 out of 33 sampling sites (Fig. 8, Table 1). Sites with erratic demagnetization directions were rejected from further analysis.

Finally, the average site-mean direction is Ds = 102.5°, Is = 62.4°, α_{95} = 3.7° (Fig. 8, Table 1) after tilt correction. The directions of

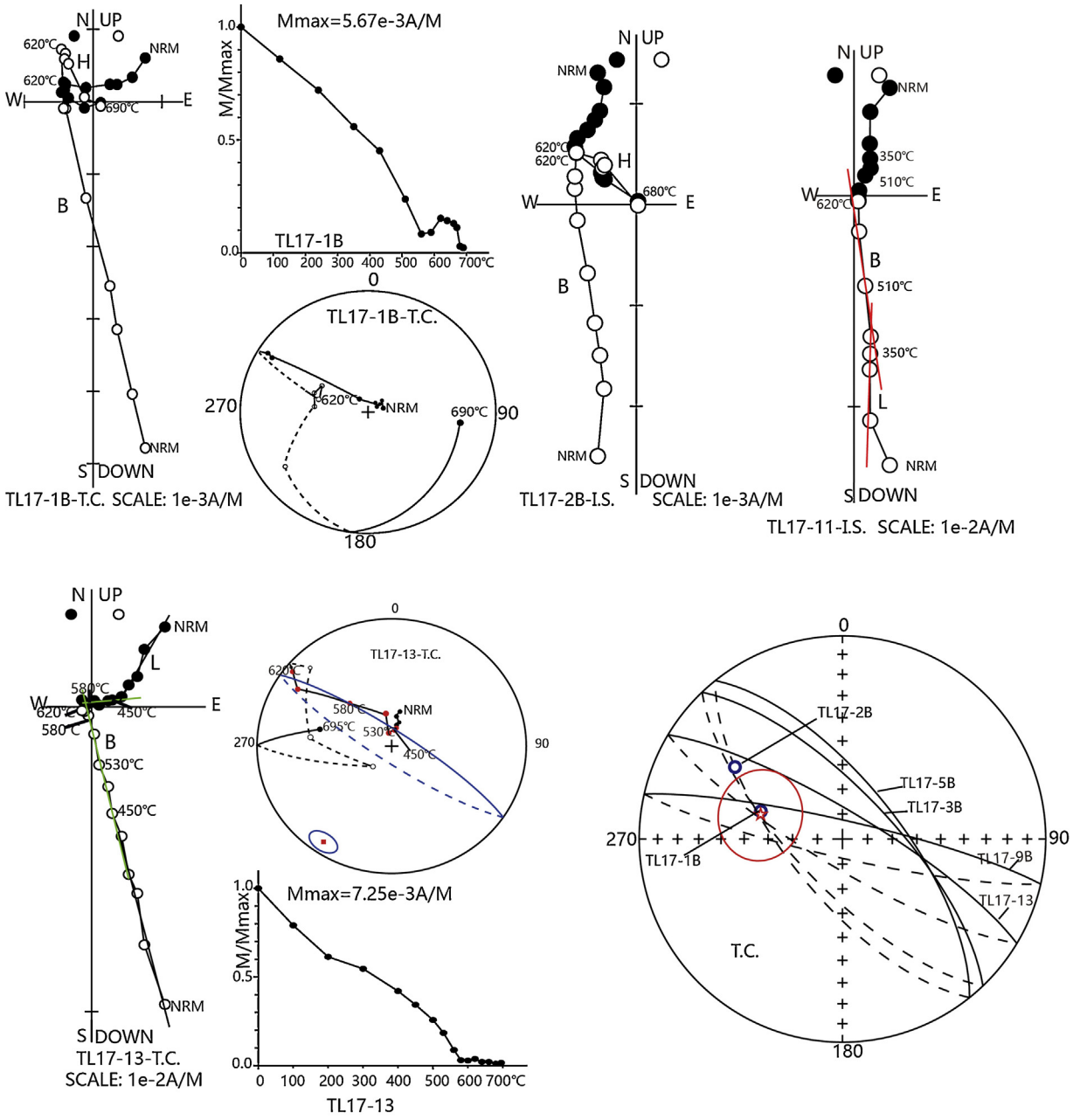


Fig. 4. Orthogonal vector projection, normalized decay curves of NRM thermal demagnetization and their stereographic projections for site 17. Samples TL17-1B and 17-2B show a linear reversed 'H' component which trends toward to the origin. Sample TL17-13 shows erratic directional changes after heating to 560–620 °C. The site TL17 mean direction (red-star) is calculated by the intersection of the great circles and from the directly observed components.

component H pass the reversal test (McFadden and Lowes, 1981) at the 95% probability level (B class).

The AMS results for the Liantuo Formation in stratigraphic coordinates demonstrate that the minimum (K3) axes are mainly perpendicular to the bedding plane, while the intermediate (K2) and maximum (K1) susceptibility axes lie within the bedding plane (Fig. 9), indicating a sedimentary origin of the AMS (Pares et al., 1999).

4.2. Wangjiapeng area (WL) (111.069° E, 30.538° N)

Three NRM components were revealed by thermal demagnetization. A component 'L' was separated below 200–400 °C in 73% of the samples (Fig. 10). The mean direction in in situ coordinates (Dg = 7.6°, Ig = 54.7°, α_{95} = 3.1°, N = 7 sites) is close to the present earth's field (PEF) (Fig. 11, Table 2), indicating a recently acquired viscous remanence. Another component, 'R', was separated mostly below 580–600 °C (Fig. 10) with a mean direction (Dg = 33.6°, Ig = 70.2°, α_{95} = 2.9°, N = 7 sites) in in situ coordinates (Fig. 11, Table 2). The corresponding pole (Lat = 56.3°, Long = 146.7°, dp/dm = 4.3°/5°) is close to the Late Triassic to Early Jurassic poles from the SCB. The negative fold test at the 95% confidence level (McFadden, 1990) suggests that the 'R' component may be attributed to remagnetization during the Yanshan Orogeny. A higher temperature component, 'H', with a blocking temperature of 690 °C can be separated (Fig. 10); however, it is too scattered to give a reliable mean direction (Fig. 12).

The AMS results for of all of the samples show that the minimum susceptibility (K3) axes range from vertical to west plunging in the stratigraphic coordinates (Fig. 13). The corresponding magnetic

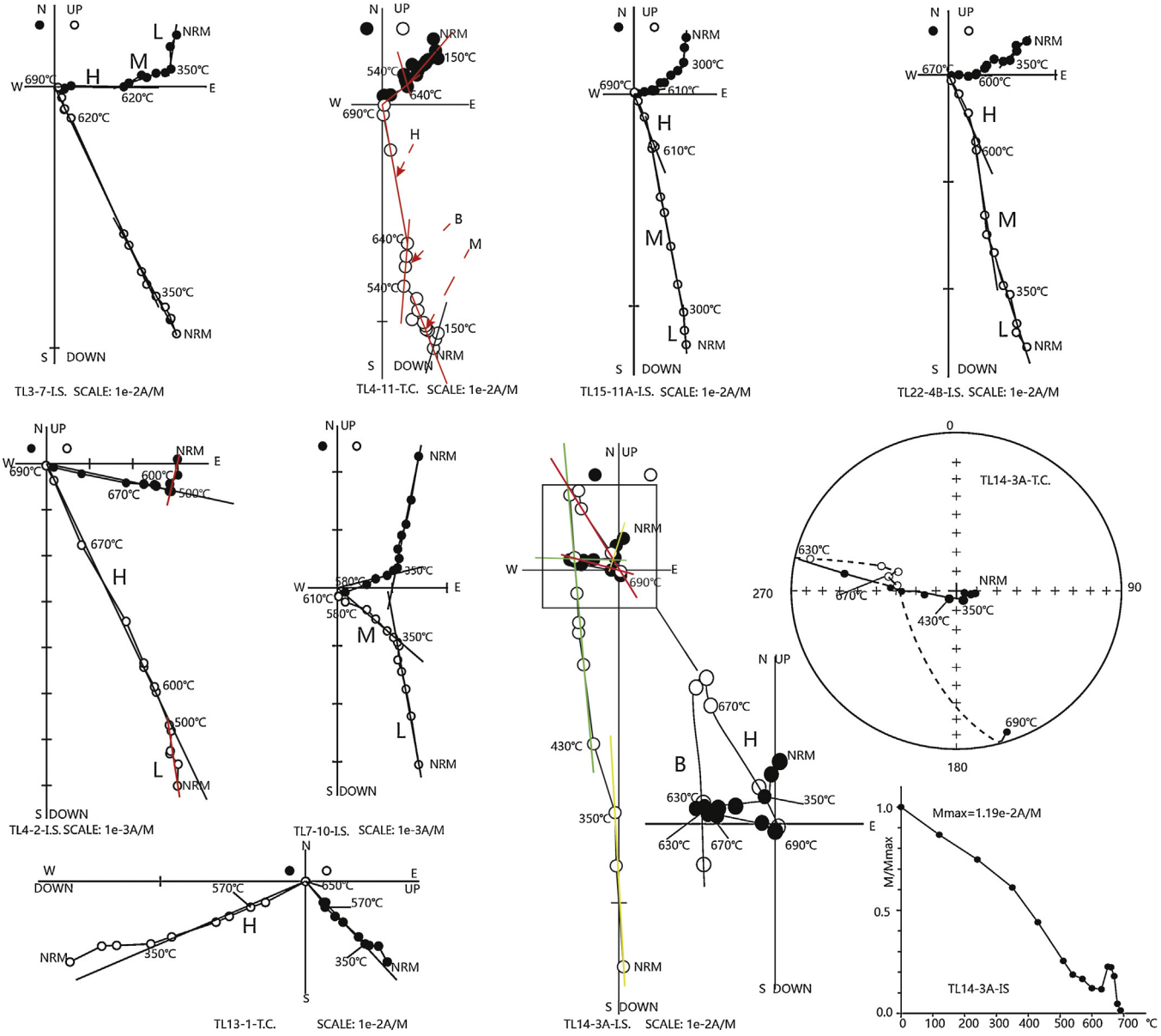


Fig. 5. Orthogonal vector projection of NRM thermal demagnetization, stereographic projection and normalized decay curve of NRM thermal demagnetization. Sample TL4-11 exhibits four components with the 'B' component after the 'M' component. Sample TL14-3A exhibits a linear reversed 'H' component trending toward the origin.

ellipsoid is prolate. This AMS pattern exhibits a 'pencil structure' which indicates a weak deformation contribution (Pares et al., 1999), which may reflect a NWW-SEE compression that led to the development of NNE-SSW thrust faults in the Wangjiapeng area during the Yanshan orogeny, as illustrated in the map of the regional geological setting (Fig. 1).

5. Discussion

5.1. Age of the Liantuo Formation and of component 'H'

The Liantuo Formation was first dated by SHRIMP U–Pb dating of twenty-three zircons separated from an interbedded tuff in the lower part of the formation; the results yielded a discordia lower intercept age of 748 ± 12 Ma (Ma et al., 1984). However, recent studies show generally younger ages from the upper part of the Liantuo Formation (Liu et al., 2008; Gao and Zhang, 2009; Hofmann et al., 2011).

Relatively young LA-ICP-MS U–Pb ages of 704 ± 6 Ma and 733 ± 7 Ma were reported by Liu et al. (2008) from tuff samples from the uppermost and middle parts of the Liantuo Formation, respectively. A mean SHRIMP zircon age of 724 Ma ± 12 Ma (n = 2, MSWD = 1.7, from two grain zircons, 714 ± 14 Ma, 735 ± 14 Ma) was obtained for the uppermost part of the Liantuo Formation, which was sampled just 60 cm beneath the Nantuo Formation (Gao and Zhang, 2009). Hofmann et al. (2011) also obtained a detrital zircon U–Pb LA-ICP-MS concordia age of 728 ± 5 Ma from a sandstone sample from the Liantuo Formation at the Huajipo section, west of the Yangtze Gorge area, in which the youngest grain had an age of ca. 710 Ma.

Recently, Lan et al. (2015) carried out a detailed SIMS U–Pb zircon study on the Liantuo Formation in the Wangfenggang and Tianjiayuanzi sections of the Three Gorges area. The results of SIMS U–Pb zircon dating of six tuff/tuffaceous siltstone beds in the Liantuo Formation suggest that it was deposited within the time interval of ca. 780–714 Ma in the Three Gorges region. In this case, we divided the Liantuo formation into lower, middle and upper

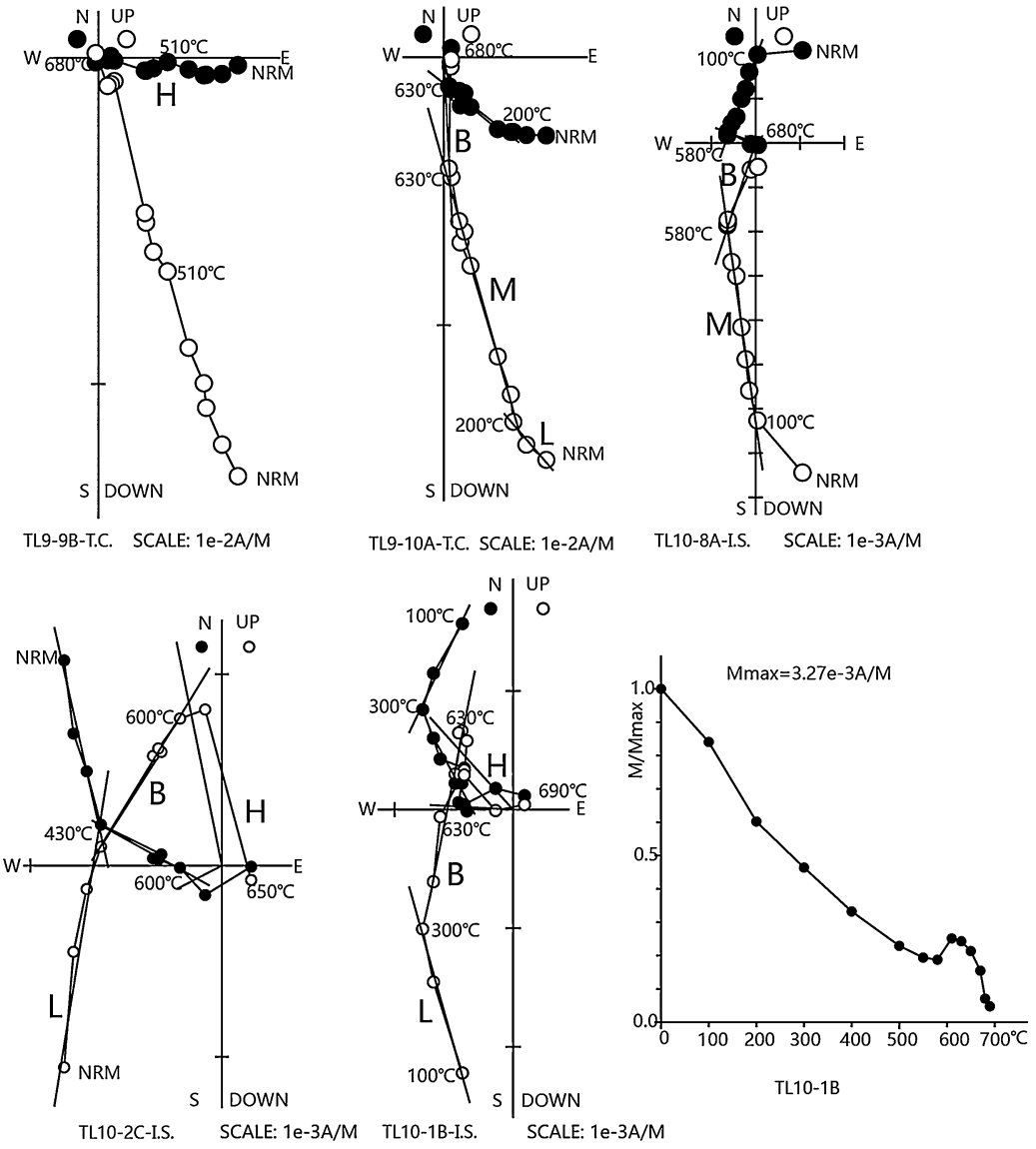


Fig. 6. Orthogonal vector projection and normalized decay curve of NRM thermal demagnetization. Samples TL10-1B,10-2C exhibit a linear reversed 'H' component which trends toward the origin. Sample TL10-8A exhibits both 'B' and 'M' components.

parts (Fig. 2). The site-mean directions of component 'H' obtained from the middle and upper parts of the Liantuo Formation are not significantly different (Fig. 14, Table 1), and therefore we give a mean direction based on all of the sampling sites. Our results reveal a polarity sequence of 5 distinct polarity chrons from the middle and upper parts of the Liantuo Formation (Fig. 2), which indicates that the new component 'H' of the Liantuo Formation is of a primary character that was probably acquired either concurrently with or shortly after deposition, and can be regarded as a reliable pole for the SCB.

It is becoming clear that the upper 20 m of the Liantuo Formation spans the time interval of ca.730 Ma–715 Ma in the Three Gorges region (see Fig. 3 in Lan et al., 2015). Considering the distribution of our sampling sites (Fig. 2), we assign the age of 720 Ma to component 'H' and its corresponding Liantuo pole. We note that the results reported by Evans et al. (2000) are also from the (middle-) upper parts of the Liantuo Formation, which could be the reason for the discordance between the Z1 pole and the MDS pole (755 Ma, Yang et al., 2004).

We suggest that component 'B', which is very similar to the 'B' pole of Evans et al. (2000), was acquired prior to the Triassic; however, the exact age of the acquisition of the remanence remains undetermined. In this case, pole B does not contain any useful information for determining the apparent polar wander path (APWP) of the South China Block.

5.2. Comparison with previous mid-Neoproterozoic paleomagnetic results

A preliminary paleomagnetic pole was reported from a few samples ($n = 14$) from the Chengjiang Formation in Yunnan province (Zhang and Piper, 1997) (Fig. 15). Although the Chengjiang Formation was considered to be correlated with the Liantuo Formation, the large age range (ca. 800–720 Ma, Jiang et al., 2012; Cui et al., 2013) is problematic. Therefore we chose to exclude the Chengjiang pole from further consideration.

Evans et al. (2000) separated a high temperature component (630–686 °C) from the middle part of the Liantuo Formation in a section 6.5 km from the Tianjiayuanzi sub-section. A comparable reversed to normal polarity sequence was revealed from three sub-sections of <10 m thickness. Comparison of the demagnetization characteristics of our results from the upper part of the formation

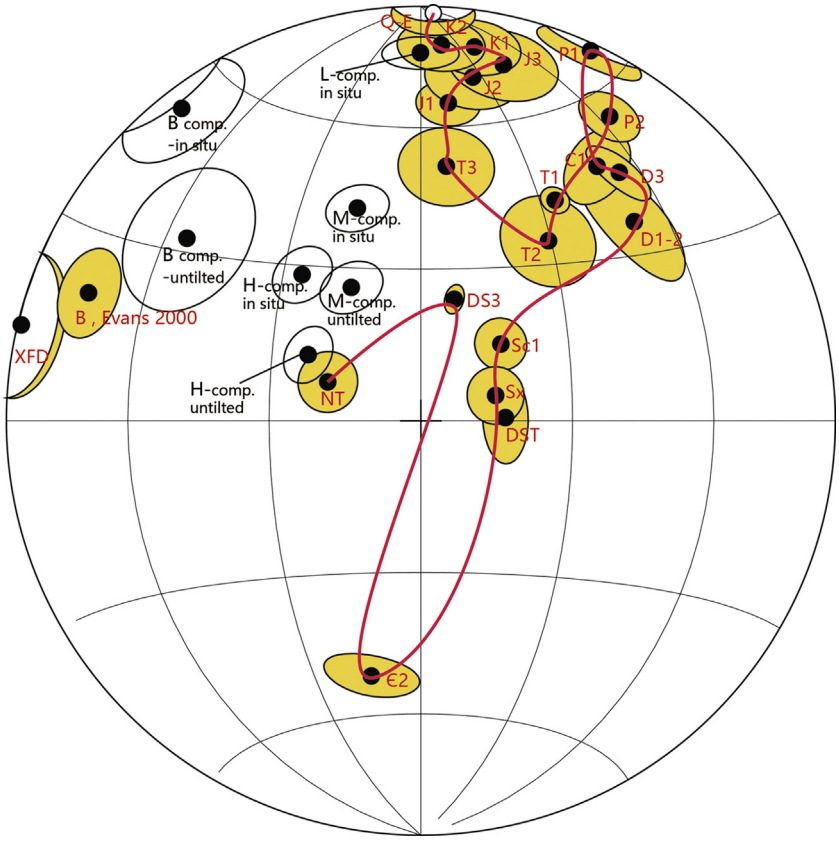


Fig. 7. Stereographic projection of paleomagnetic poles calculated from components 'L', 'M', 'B', and 'H'. The existing Neoproterozoic and Phanerozoic paleopoles (gray yellow shadow) from the South China block are also shown for comparison. The XFD pole reported by Li et al. (2004) is very close to the 'B' component pole obtained by Evans et al. (2000). NT = Nantuo Fm pole (Zhang et al., 2013), DS3 = Doushantuo Fm Member3 pole (Zhang et al., 2015), Cam2 = middle Cambrian, O1-2 = early-middle Ordovician, S2-3 = middle-late Silurian, D1-2 = early-middle Devonian, D3 = late Devonian, C1 = early Carboniferous, P1 = early Permian, P2 = late Permian, T1 = early Triassic, T2 = middle Triassic, T3 = late Triassic, J1 = early Jurassic, J2 = middle Jurassic, J3 = late Jurassic, K1 = early Cretaceous, K2 = late Cretaceous, Q-E = Cenozoic.

(sites TL10 and TL14, Fig. 2) with results obtained by the CIT group (Evans et al., 2000) reveals almost identical behavior: a composite of a steeper component 'B' and a similar high temperature component with reversed polarity. Thus, the mean directions from three sub-sections are considered as four independent sites (CIT lower, CIT upper, UWA site 1, 2, in their Table 1) yielded by the CIT and UWA groups (Evans et al., 2000), which are used to average with our results. The averaged direction is Ds = 103.4°, Is = 64.1°, α95 = 3.7°, yielding a grand mean pole for the upper Liantuo Formation at 13.2° N, 155.2° E with A_{95} = 5.3°.

Table 2
Site-mean directions calculated from the palaeomagnetic results of different temperature components of samples from the Liantuo Formation at Changyang, South China (Mean GPS: 111.069° E, 30.538° N).

Site	Comp.	n/N	Dg	Ig	Ds	Is	k	α95
WL1	L	7/10	6.2	58.1	355	57.6	37	10.1
	R	8/10	31.8	69.6	12.7	71.7	104.8	5.4
WL2	L	9/11	5.7	56.3	359.4	49.9	23.4	10.9
	R	10/11	28.4	72.4	11.7	67.2	170.9	3.7
WL3	L	10/11	14.3	53.8	7.4	48.1	79.3	5.5
	R	10/11	45.5	70.7	25.6	67.8	470.9	2.2
WL4	L	4/10	358.4	51	354.8	52.1	80.6	10.3
	R	7/10	23.8	64.9	17.9	66	59.7	7.9
WL5	L	9/10	11.5	52.3	2.8	51.2	69.7	6.2
	R	7/10	27.9	70.3	8.1	70.6	98.4	6.1
WL6	L	5/10	13.1	55.2	11.2	51.5	46.4	11.4
	R	9/10	44.4	69.5	36.7	66.9	90.7	5.4
WL7	L	7/9	4.2	55.3	1	48.7	31.7	10.9
	R	6/9	35.7	72.8	21.3	68	162.2	5.3
Mean R components								
		7/7	33.6	70.2			430.1	2.9
					19.5	68.6	379.6	3.1
Paleomagnetic pole:				56.3 N, 146.7 E, dp/dm = 4.3/5 (In situ)				
Mean L components								
		7/7	7.6	54.7			386.6	3.1
Paleomagnetic pole:				79 N, 175.8 E, dp/dm = 3.5/4.9 (In situ)				

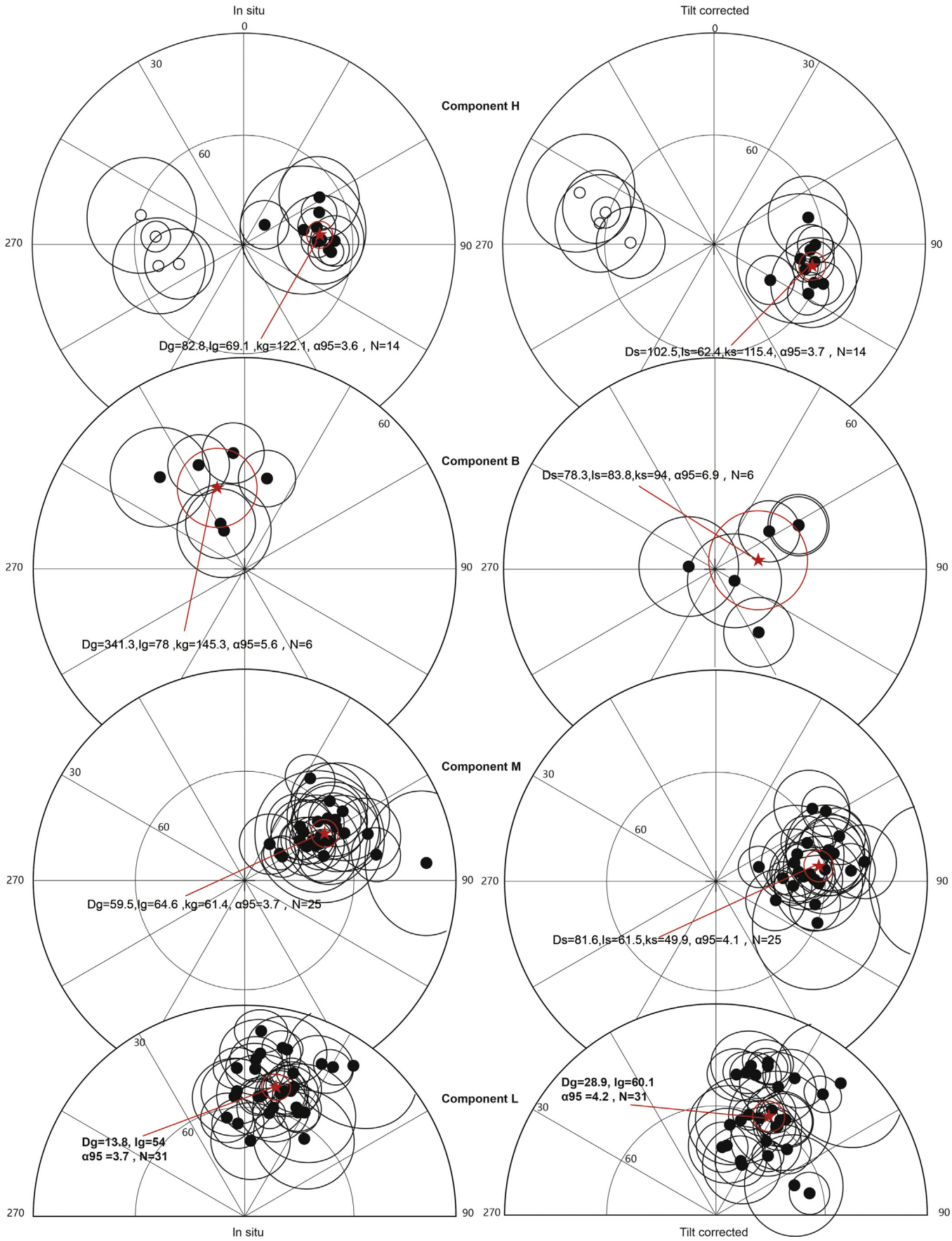


Fig. 8. Equal-area stereographic projections of site-mean directions of different components ('L', 'M', 'B' and 'H') from Tianjiayuanzi sections in situ (I.S.) and after tilt correction (T.C.). Red stars indicate the site-mean directions. Lower (upper) hemisphere directions are indicated by closed (open) symbols.

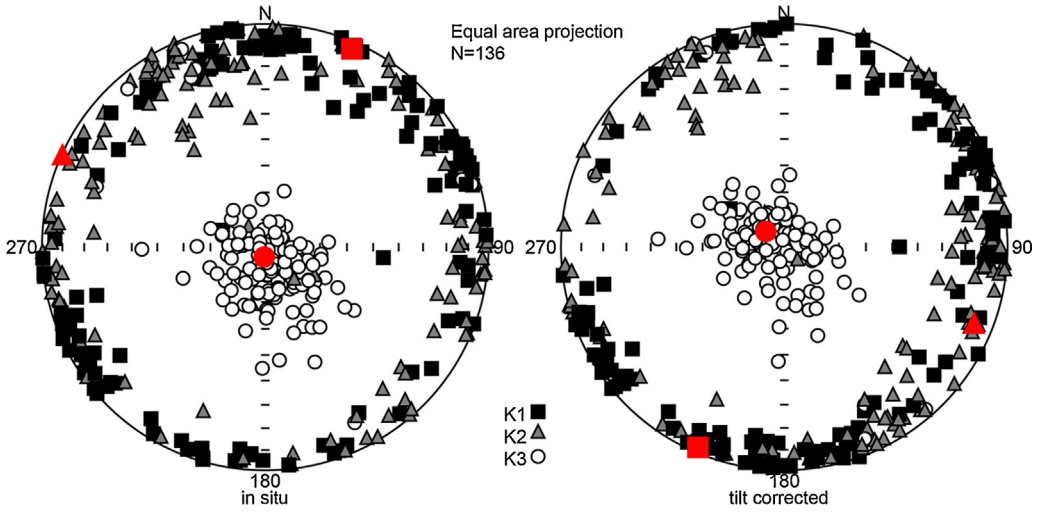


Fig. 9. Equal area projections of principal susceptibility axis directions for the Tianjiayuanzi sections. Red circles, triangles and squares indicate mean minimum, intermediate and maximum principal susceptibility axes, respectively.

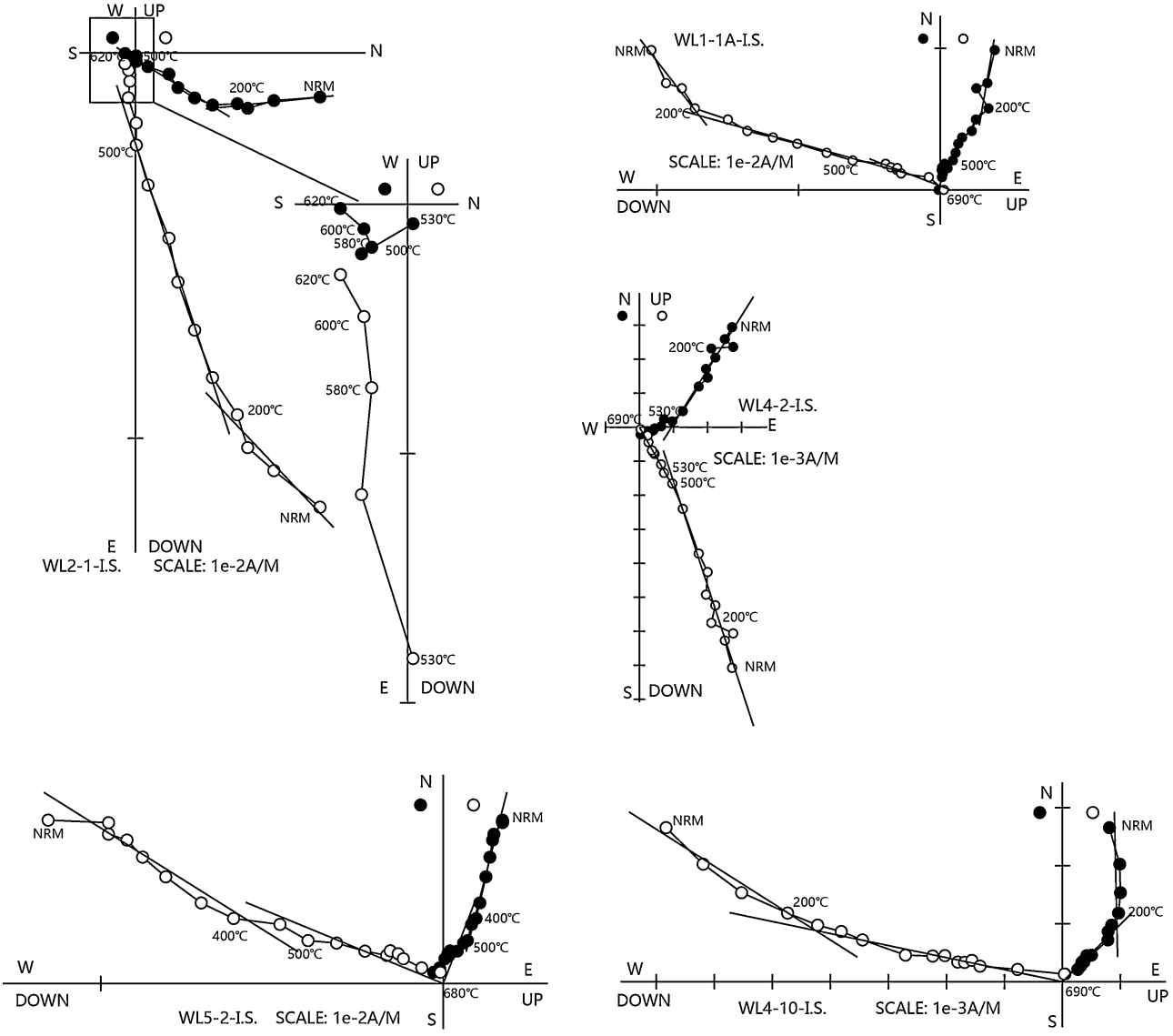


Fig. 10. Orthogonal vector projection of NRM thermal demagnetization from the Wangjiapeng section in in situ coordinates. Note that the 'L' and 'R' components are removed below 400 °C. The 'H' components are separated until 690 °C.

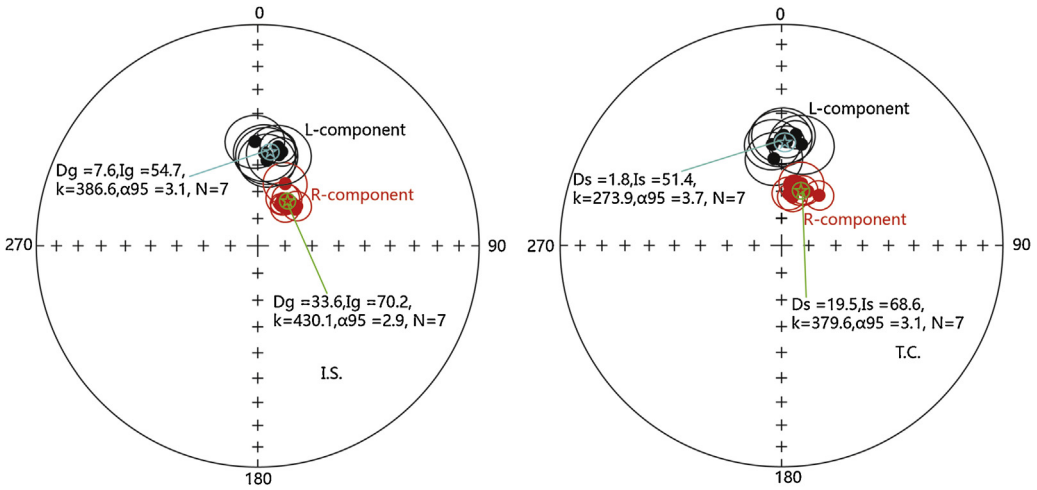


Fig. 11. Equal-area stereographic projections of site-mean directions both in in situ (I.S.) and after tilt correction (T.C.). Light-blue and green open stars indicate the site-mean directions of the 'L' and 'R' components, respectively.

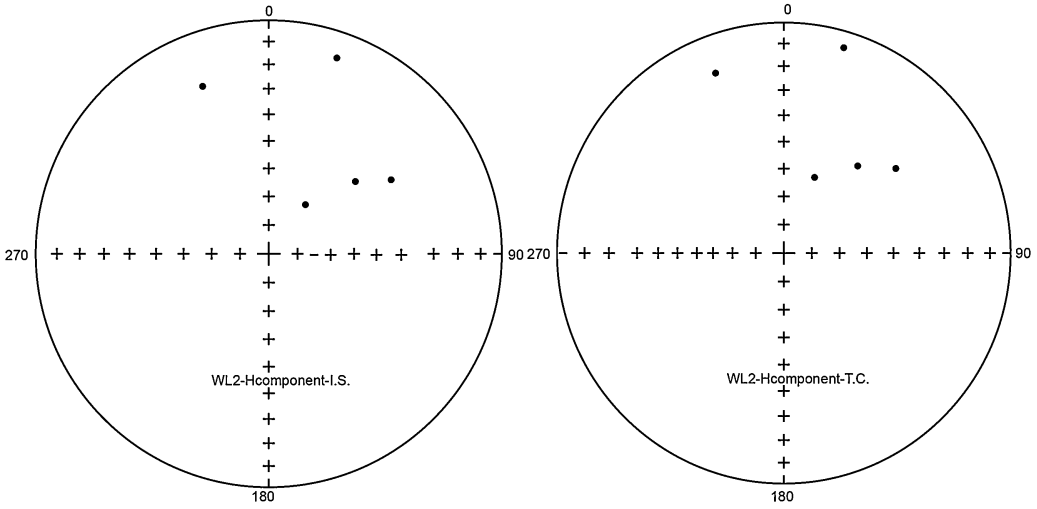


Fig. 12. Equal-area stereographic projections of the 'H' components of site WL2 both in situ (I.S.) and after tilt correction (T.C.). The directions of the 'H' component are too scattered for reliable calculation of a mean direction.

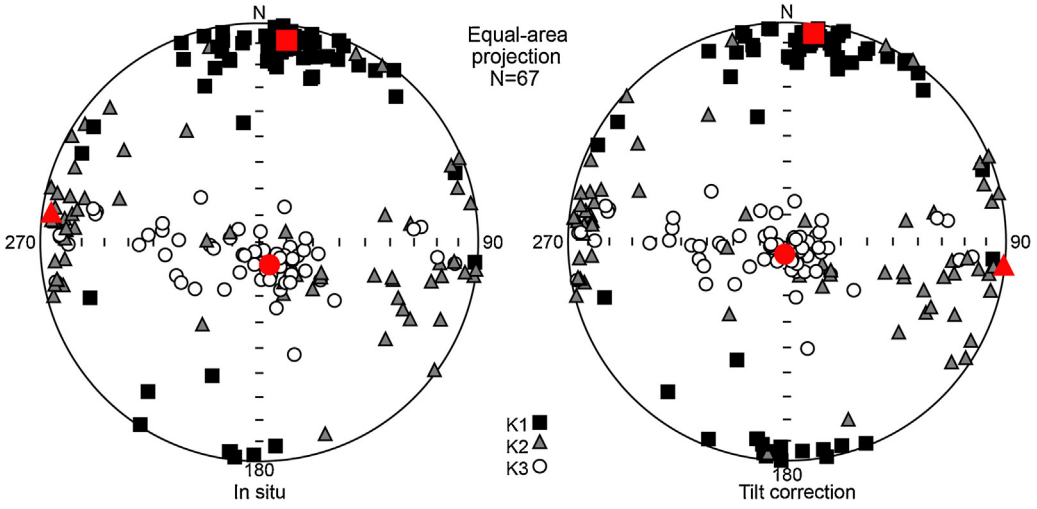


Fig. 13. Equal area projections of principal susceptibility axis directions for the Wangjiapeng section. Red circles, triangles and squares indicate mean minimum, intermediate and maximum principal susceptibility axes, respectively.

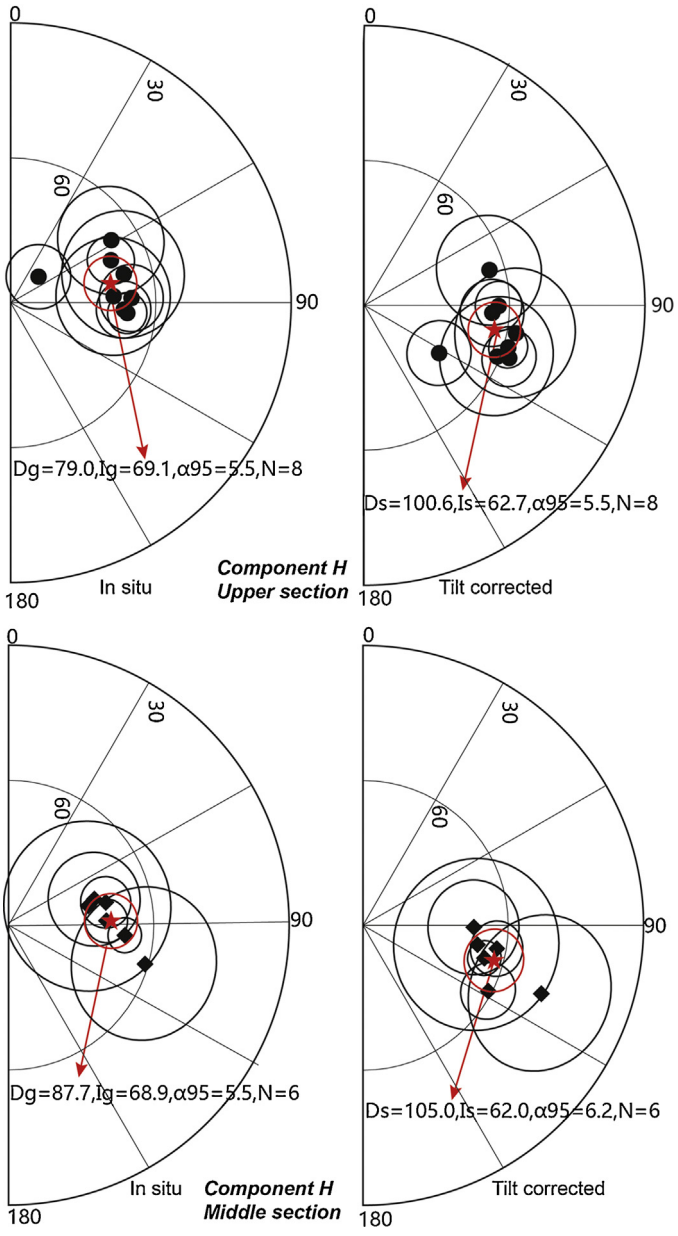


Fig. 14. Comparison of the site-mean directions of 'H' components obtained from the upper and middle parts of the Liantuo Formation at Tianjiayuanzi sections in situ (I.S.) and after tilt correction (T.C.). Red stars denote the mean directions.

5.3. Paleogeographic implications

Recently, Schmidt (2014) reviewed the Precambrian paleomagnetic results for Australia. From the dataset, seven middle and late Neo-Proterozoic paleopoles obtained from South and West Australian cratons with Q > 4 were selected and used to compare with coeval poles of the SCB (Table 3). Combined with the paleomagnetic results of Embleton and Schmidt (1985) from the Northampton inlier, Wingate and Giddings (2000) obtained a high quality paleomagnetic pole with a U–Pb baddeleyite age of 755 ± 3 Ma from the Mundine Dyke Swarm (MDS) which passes a convincing contact test. The ca. 640 Ma pole of the Yaltipena formation (YF) (Sohl et al., 1999) and the ca. 635Ma pole of the Elatina Formation (GEF) (Schmidt et al., 2009) derived from the central Flinders Ranges, both pass the fold and reversal tests. This evidence demonstrates that they preserve early-acquired remanent magnetizations.

Schmidt and Williams (1996, 2010, 2013) and Schmidt et al. (2009) carried out a series of paleomagnetic studies of the Australian Ediacaran stratigraphy, including the Nuccaleena Formation (NF), the Bunyeroo Formation (BU), the Brachina Formation (Bra) and the Wonoka Formation (WF). All of the paleomagnetic data from these formations demonstrated positive fold tests and dual polarities. In addition, their ChRM directions are different to each other, indicating the primary character of the remanences. However, as demonstrated in Fig. 15 (data listed in Table 3), these paleopoles exhibit a large change along a small circle centered in the sampling areas of the Adelaide Geosyncline. Within the polar wander path for 640–540 Ma, there are two rapid swings, especially in an easterly direction from 590 to 560 Ma, followed by a return in a westerly direction from 560 to 545 Ma. It is impossible for two large block rotation events of East Gondwana to have occurred before the Pan-African Orogeny at around the almost constant paleolatitude of 23° S of the sampling areas during the interval of 590–545 Ma. Although Schmidt (2014) claimed that the inclination shallowing had occurred in the argillaceous rocks of the Brachina, Bunyeroo and Wonoka Formations, this cannot explain the oscillations of the late Ediacaran polar wander. Clearly, more data are needed to exclude the possibility of local rotations that took place within the Adelaide fold belts.

Because the ca. 800–760 Ma Hussar pole reported by Pisarevsky et al. (2007) is similar to the late Devonian Retreat batholith pole (Anderson et al., 2004), and lacks a field test and exact dating, we have chosen to abandon it. For the early Paleozoic, we use the Gondwana grand mean poles summarized by McElhinny et al. (2003) (listed in their Table 3), which are rotated back to Australian coordinates around a Euler pole at 28.1° S, 293.2° E with an angle of −52.1° (McElhinny et al., 2003).

Two Ediacaran paleopoles from the Doushantuo Formation were reported from south China, including one from Hunan province (Macouin et al., 2004) and another from the Three Gorges area of Hubei province (Zhang et al., 2015). Because the pole from Hunan is close to the Silurian poles reported by Huang et al. (2000) and Opdyke et al. (1987), Zhang et al. (2015) suggested that the result was a Silurian remagnetized remanence. However, the new paleomagnetic data from Member 3 of the Doushantuo Formation (14 m below Member 4), with reversed and normal polarities, exhibit an interesting systematic trend in inclination through the sampling strata (Zhang et al., 2015). The inclinations varied by more than 45°in two layers covering less than 30 cm thickness of limestone (Sites JLW06-JLW09, and ZG04-JLW11 in Fig. 2 of Zhang et al., 2015). Four possible explanations, including differing degrees of unresolved secondary contamination of the HTC remanence, have been proposed and were excluded by the authors. We plot all of the site-mean paleopoles reported by Macouin et al. (2004) and Zhang et al. (2015), and those for the Silurian reported by Huang et al. (2000) and Opdyke et al. (1987), in Fig. 16A. It is interesting to note that if the results from Hunan province were remagnetized during the Silurian, then the possibility cannot be excluded that some of the sampling rocks of the Doushantuo Formation from the Three Gorge area were also remagnetized at the same time (Fig. 16A). In fact, their results can be fitted to two great circles (Fig. 16B), in which one great circle comprises the directions of their soft component (SC) (Zhang et al., 2015) and the expected direction calculated from the Silurian pole (Huang et al., 2000); and the other great circle comprises the Silurian directions (Huang et al., 2000) and all of the ChRM directions of Zhang et al. (2015). It seems that the ChRM directions are clustered around those for the Silurian, or along the great circle distribution, suggesting the effects of remagnetization of the Doushantuo Formation during the Silurian. However, some of the remanences at the terminus of the great circle may perhaps be interpreted as a primary magnetization. Therefore, we tentatively calculate a mean pole using the data

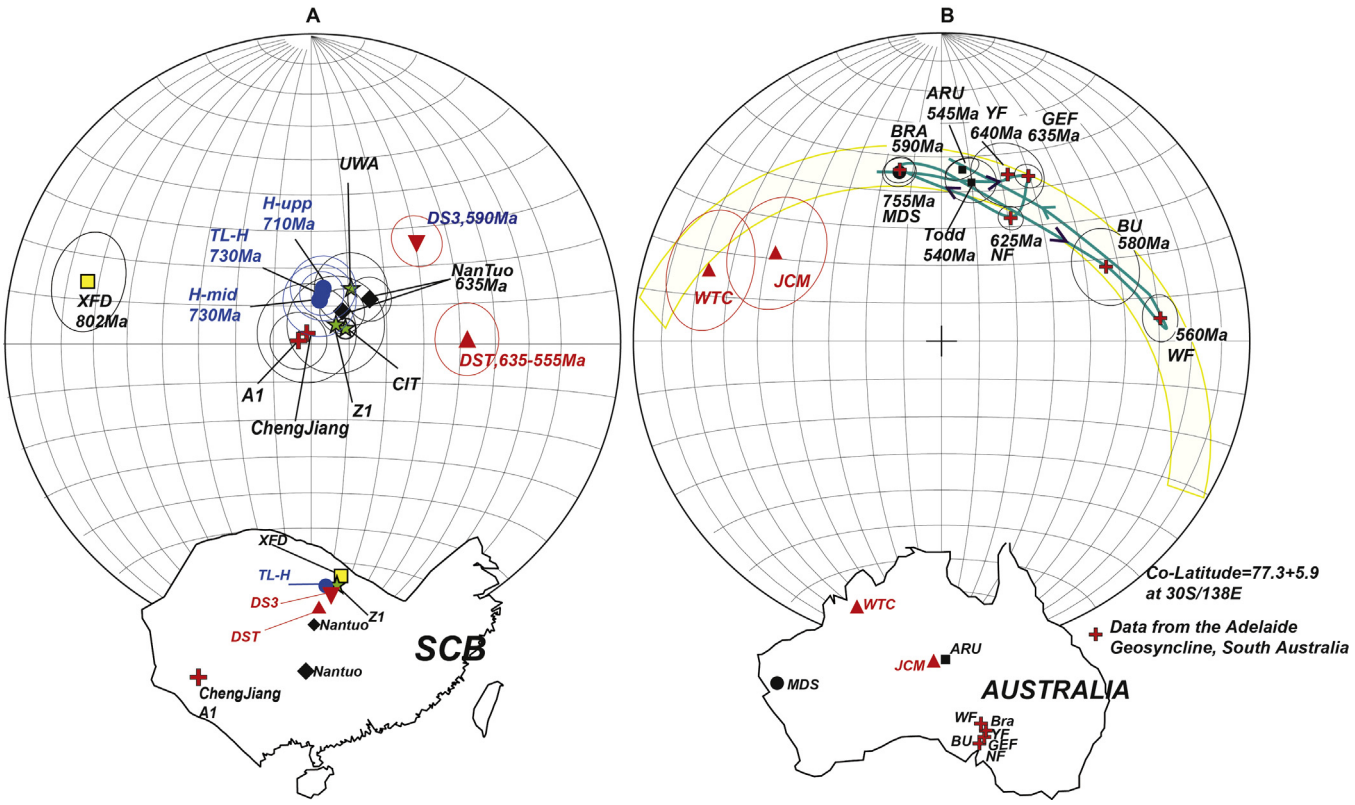


Fig. 15. (A) Comparison of poles calculated from the 'H' component (this study) with previous late Neoproterozoic poles and their sampling site distributions from the South China block. XFD (square) denotes the Xiaofeng Dyke pole (Li et al., 2004); The Z1, CIT and UWA (star) poles indicate the mean pole of the Liantuo Formation reported by Evans et al. (2000), and obtained by the CIT and UWA groups, respectively. The A1 and Chengjiang (crosses) poles are from the Nantuo Formation and Chengjiang Formation, respectively (Zhang and Piper, 1997). The Nantuo pole (diamond) represents results from the Nantuo Formation (Zhang et al., 2013). The DS3 (Inverted triangle) and DST pole (triangle) represent results from the Doushantuo Formation (Zhang et al., 2015; Macouin et al., 2004), respectively. The poles calculated from the TL-H (all sites), H-upp (upper part), H-mid (middle part) components of the Liantuo Formation are shown for comparison. (B) Selected late Cryogenian-Early Cambrian poles of Australia (listed in Table 3) and their sampling site distributions. These paleopoles exhibit a rapid change along a small circle centered in the sampling areas in the Adelaide Geosyncline.

Table 3
Summary of Late Cryogenian-Early Paleozoic palaeomagnetic poles derived from the Australia & South China Blocks (with quality index Q > 4).

Pole name	Formation	Age	1	2	3	4	5	6	7	Q	Plat N	Plong E	A95	References
Australia														
MDS	Mundine well dyke swarm	755 ± 3 Ma	√	√	√	√	√	n	n	5	45.3	135.4	4.1	Wingate and Giddings, 2000
YF	Yaltipena Fm	640 Ma?	n	√	√	√	√	√	n	5	44.2	172.7	9	Sohl et al., 1999
GEF	Elatina Fm	635 Ma?	n	√	√	√	√	√	n	5	43.7	179.3	3.3	Schmidt et al., 2009
NF	Nuccaleena Fm	625 Ma?	n	√	√	√	√	√	√	6	32.3	170.8	3.2	Schmidt et al., 2009
Bra	Brachina Fm	609 ± 64 Ma	n	√	√	√	√	√	√	6	46	135.4	3.7	Schmidt and Williams, 2010
BU	Bunyeroo Fm	593 ± 32 Ma	n	√	√	√	√	√	n	5	18.1	196.3	9.5	Schmidt and Williams, 1996
WF	Wonoka Fm	556 ± 24 Ma	n	√	√	√	√	√	n	5	5.2	210.5	5.4	Schmidt and Williams, 2010
South China block														
TL-H	Liantuo Fm	720 ± 15 Ma	√	√	√	n	√	√	√	6	12.7	157.4	5.2	This study
Z1	Liantuo Fm	720 ± 15 Ma	√	√	√	n	√	√	√	6	4.1	161.1	12.9	Evans et al., 2000
LT	Average pole for Liantuo Fm		√	√	√	n	√	√	√	6	13.2	155.2	5.27	This study
A1	Nantuo Fm	635.2 ± 0.6 Ma	n	√	√	√	√	√	√	6	0.2	151.2	6.5	Zhang and Piper, 1997
NT	Nantuo Fm	635.2 ± 0.6 Ma	√	√	√	√	√	√	√	7	9.3	165.9	4.3	Zhang et al., 2013
DST	Doushantuo Fm	635–550 Ma	√	√	√	√	√	n	n	5	0.6	196.9	7.1	Macouin et al., 2004
DS3	Doushantuo Fm	595 ± 22 Ma	√	√	√	n	√	√	n	5	25.9	185.5	6.7	Zhang et al., 2015
cam2	Douposi Fm	510 ± 11 Ma	√	√	√	√	√	√	√	7	−51.3	166	6.8	Yang et al., 2004
O3	Pagoda Fm	458 ± 6 Ma	√	√	√	√	√	√	√	7	−45.8	191.3	3.2	Han et al., 2015
Sx	Xiushan Fm	430 ± 2 Ma	√	√	√	√	√	n	√	6	4.9	194.7	5.6	Opdyke et al., 1987
Sc1	Rongxi, Huixingshao, Niugundang Fm	430 ± 7 Ma	√	√	√	√	√	n	√	6	14.9	196.1	5.1	Huang et al., 2000

1–7 and Q, reliability criteria of Van der Voo (1990). 1 – Well-dated rock age; 2 – Sufficient number of samples; 3 – Adequate demagnetization; 4 – Field tests; 5 – Structural control and tectonic coherence with craton or block involved; 6 – The presence of reversals; 7 – No resemblance to paleopoles of younger age. Detail discussion on the age of Australian Ediacaran Formations was presented by Schmidt and Williams (2010).

from sites ZG2, 3, 4 (Zhang et al., 2015), representing the pole of the Doushantuo Formation (DSTM3: 32.3° N, 161.4° E, A_{95} = 9.06° before tilt correction, and 32.7° N, 170.1° E, A_{95} = 8.09° after tilt correction).

The APWP was established using the key higher quality poles available from the Yangtze craton in south China, including the Liantuo pole (LT) from the Three Gorges area (720 Ma, Evans et al., 2000; this study), the Nantuo Formation pole (635 Ma, Zhang et al.,

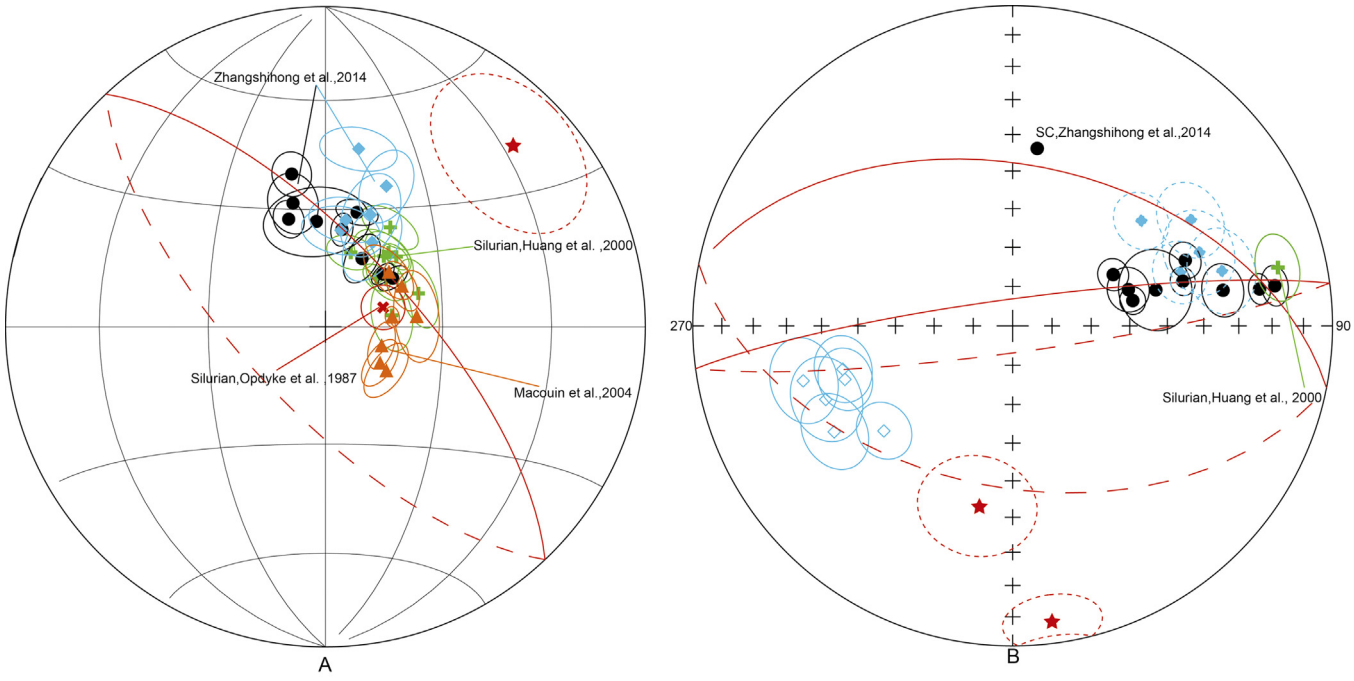


Fig. 16. (A) Equal-area stereographic projections of reported paleomagnetic poles of the Doushantuo Formation (Zhang et al., 2015; Macouin et al., 2004) and the Silurian poles (Opdyke et al., 1987; Huang et al., 2000), the poles calculated from each site of Zhang et al. (2015) are distributed along a great circle; (B) equal-area stereographic plot of the site-mean directions reported by Zhang et al. (2015), to which two great circles can be fitted. SC = secondary component. The red stars with errors are the pole of the fitted great circles.

2013), the recalculated Doushantuo pole (590–560 Ma, Zhu et al., 2013), the middle Cambrian pole (Yang et al., 2004), a late Ordovician pole that passes the reversal and fold tests (45.8° S, 191.3° E, dp = 2.4°, dm = 4.2°) from the northern part of the Sichuan basin (Han et al., 2015) and the Silurian poles (Huang et al., 2000; Opdyke et al., 1987).

A comparison can be made of coeval poles for South China and Australia, including the Nantuo (NT, 635 Ma), DSTM3 (590–560 Ma) and the middle Cambrian (Cam2) poles of South China and the counterparts YF, GEF, Bra and the ca. 520 Ma poles of Australia (Fig. 17A). By rotating the SCB and its APWP around an Euler pole at 28° N and 142.1° E with angle = 70°, we can reconstruct the South China block with respect to West and South Australia during the period of 635–520 Ma, which places South China to the northwest of Northern Australia.

Although Zhang et al. (2015) preferred the correlation of the Doushantuo Formation with the Wonoka Formation, the uncertain age of the Ediacaran strata in the Adelaide Geosyncline (Calver et al., 2004; Kendall et al., 2006) implies that the comparison is preliminary and further work is needed to substantiate it. In addition, because the ~720 Ma LT pole is distinct from the MDS, direct comparison of the two may lead to an incorrect reconstruction of the SCB relative to Australia during the Neoproterozoic. Nevertheless, the paleolatitude of the Yangtze Block at 720 Ma (39.7 ± 6.3°, reference point: 23° N, 103° E) is close to the paleolatitudinal trend line of Northern Australia calculated from ~755 Ma to ~640 Ma poles in northern Australia coordinates (37.1 ± 4.5° and 24.1 ± 7.5°, respectively, at reference point: 15° S, 120° E), implying that the South China block was close to the northwest of Northern Australia at ~720 Ma, or earlier.

In contrast to the debated relationship between south China and Australia during the Neoproterozoic, there is a general consensus regarding the Early Paleozoic paleogeographic reconstruction between the SCB and Australia (Yang et al., 2004; Macouin et al., 2004; Zhang et al., 2013; Li et al., 2013). By rotating the SCB and its early Paleozoic APWP around an Euler pole at 10.4° N and 140° E with angle = 71°, the SCB is placed to the northwest of the Western Australia for the period from the middle Cambrian to the Silurian (Fig. 17B). A similar change of the paleoposition of the SCB from a position northwest of Northern Australia to one to the northwest of Western Australia can be also found in Li et al. (2013); however, our reconstructions are supported by the paleomagnetic results. Although Li and Evans (2011) have proposed that the Northern Australian craton was rotated about 40° counter-clockwise with respected to the Western and Southern Australian cratons along a megashear zone in the Paterson and Petermann orogens during the late Neoproterozoic (650–550 Ma), the displacement of the SCB may have occurred through the counter-clockwise rotation of the Northern Australian craton, but at the later time of 550 Ma or during the middle Cambrian. The displacement may correlate to the reactivation of the Darling fault during 550–500 Ma (Harris and Li, 1995; Compston and Arriens, 1968), which Powell and Pisarevsky (2002) have used to explain the collision of the India and Western Australia during the assembly of East Gondwana.

Based on a compilation of the spatiotemporal evolution of the Late Neoproterozoic–Early Paleozoic strata and their depositional environments, Jiang et al. (2014) proposed a new model of a retroarc thrust wedge/foreland basin system of the SCB during the Neoproterozoic–Early Paleozoic period. Two major cycles of evolution of the retroarc thrust wedge/foreland basin system occurred during <725–<635 Ma and ca. 564–446 Ma, during which two intervals of northwestward subduction of the oceanic slab beneath the southern margin of the SCB are recorded. Terrane accretion was inferred to have taken place during the late stage of these cycles. The change of the paleoposition of the South China block in the early-middle Cambrian is represented by the early Cambrian orogenic loading (Fig. 8 in Jiang et al., 2014). The paleogeographic relationship between the SCB and western Australia during the Cambrian is also supported by U–Pb ages and Hf isotope data from detrital zircons from the Cambrian sequence of the southern part of SCB, which defines two distinctive age distributions which peak at 1120 Ma and 960 Ma, respectively. The εHf (t) values of each

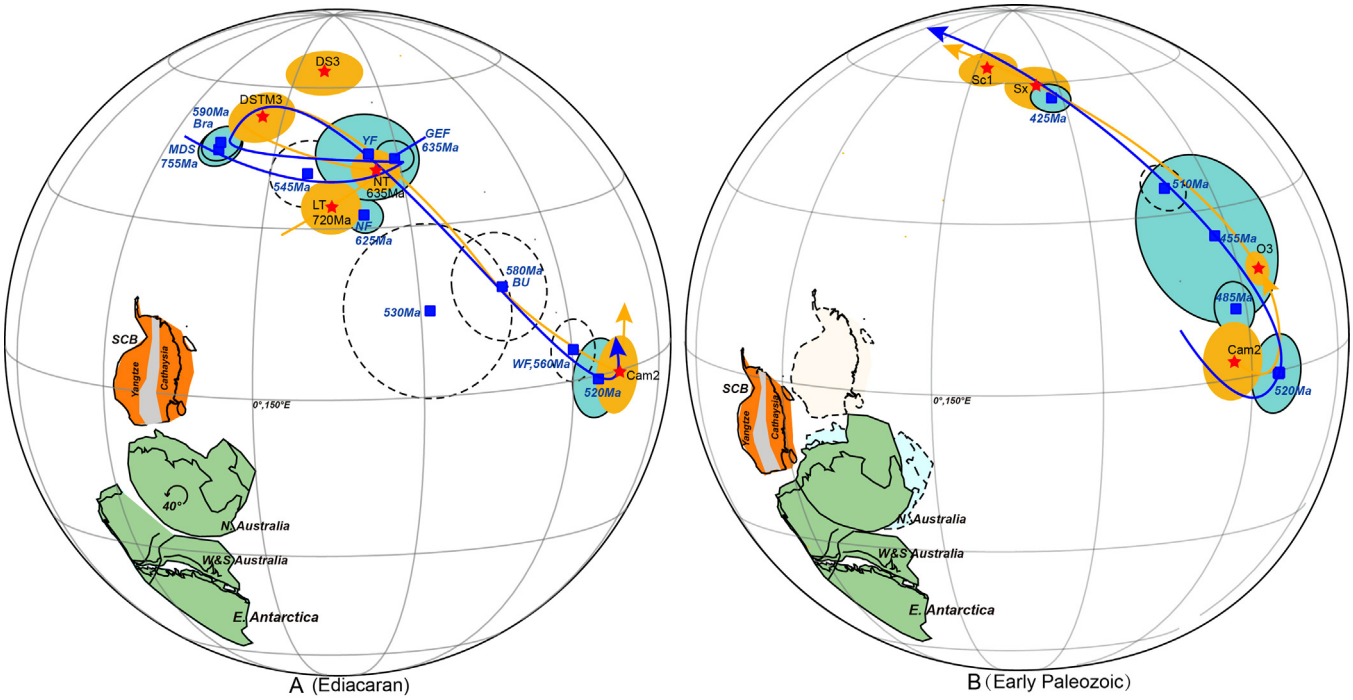


Fig. 17. Paleomagnetically based paleogeographic reconstruction of South China and Australia & Mawson cratons for the late Neoproterozoic and early Paleozoic in South & West Australian coordinates. (A) The late Neoproterozoic paleopoles (yellow) of the SCB are rotated by 70° relative the South & West Australian cratons using an Euler pole at 28° N, 142.1° E, which places the South China proximal to the northwest of northern Australia; (B) comparison of Early Paleozoic poles using an Euler rotation (10.4° N, 140° E, 71.2°) which restores the South China block to the northwest of Northwestern Australia. The Northern Australian craton is restored to South & West Australian coordinates using an Euler rotation (−20° N, 135° E, 40°), following Li and Evans (2011), during the late Ediacaran. Data are listed in Table 3. SCB = South China block, N. Australia = Northern Australia, W&S Australia = Western & Southern Australia, E. Antarctica = eastern Antarctica.

group are identical to the coeval detrital zircons from Western Australia and the Tethyan Himalaya zone (Xu et al., 2013). The authors suggested that the ca. 1120 Ma detrital zircons were most likely derived from the Wilkes-Albany-Fraser belt between southwest Australia and Antarctica. Faunal affinities of Early Paleozoic material in south China and Australia also indicate a close link with coeval strata along the northern margin of East Gondwana (Metcalfe, 2006; Hughes et al., 2005; McKenzie et al., 2011).

Finally, in reevaluating Cryogenian and Ediacaran paleomagnetic poles from Australia, we recognize a problem in that the observed strong oscillation of late Ediacaran poles is seemingly contradictory with plate movements. The overlap between the MDS pole and the Brachina pole (ca.590 Ma, Schmidt and Williams, 2010) is either a late remagnetization of the MDS pole or a simple inclination shallowing (Schmidt, 2014), or a polar wander that needs to be confirmed. In addition, the ca.100 Ma gap (ca.730 Ma–ca. 635 Ma) between poles which existed in both south China and Australia (the gap between the MDS pole (755 Ma) and the YF pole (ca. 645 Ma)) needs to be filled. The Ediacaran paleogeographic reconstruction between south China and Australia should be tested by new paleomagnetic and stratigraphic dating results from both blocks.

6. Conclusions

This study presents a more reliable Neoproterozoic pole obtained from the northwestern part of the Yangtze Block. Comparison of the results from the middle and upper parts of Liantuo Formation demonstrates that there is no significant difference between these two parts. The primary nature of the magnetization is demonstrated by its thermal stability, and by the definition of a polarity sequence with at least 5 polarity chrons.

Comparison of the APWPs of South China and Australia suggests that the South China block was located to the northwest of Northern Australia during the Ediacaran period, and may then have moved to the northwest of Northwestern Australia through counter-clockwise rotation of the Northern Australian craton at the later time of 550 Ma or in the middle Cambrian.

Reevaluation of the Cryogenian paleomagnetic poles of Australia and south China reveals that there is a ca. 100 Ma gap between the poles, spanning the interval ca. 730–635 Ma, which exists in both south China and Australia. In addition, the strong oscillation of the Australian late Ediacaran poles seems to be incompatible with plate movements. Therefore, more work needs to be done both in south China and Australia to resolve these problems.

Acknowledgements

We gratefully acknowledge the careful and constructive comments of David A.D. Evans, S.A. Pisarevsky and two anonymous viewers which considerably improved the manuscript. We are particularly grateful to Dr. Pengju Liu for guidance in the field. This work was funded by the Chinese National Natural Science Foundation (Grant 41230208).

References

Anderson, K.L., Lackie, M.A., Clark, D.A., 2004. Palaeomagnetic results from the Palaeozoic basement of the southern Drummond Basin, central Queensland, Australia. Geophys. J. Int. 159, 473–485.
Calver, C.R., Black, L.P., Everard, J.L., Seymour, D.B., 2004. U-Pb zircon age constraints on late Neoproterozoic glaciation in Tasmania. Geology 32, 893–896.
Cogne, J.P., 2003. PaleoMac: a Macintosh (TM) application for treating paleomagnetic data and making plate reconstructions. Geochem. Geophys. Geosyst., 4.
Compston, W., Arriens, P.A., 1968. The Precambrian geochronology of Australia. Can. J. Earth Sci. 5, 23.

Condon, D., Zhu, M.Y., Bowring, S., Wang, W., Yang, A.H., Jin, Y.G., 2005. U-Pb ages from the Neoproterozoic Doushantuo Formation, China. Science 308, 95–98.
Cui, X.Z., Wang, J.X.S.J, et al., 2013. Zircon U-Pb geochronology for the stratotype section of the Neoproterozoic Chengjiang Formation in Central Yunnan and its geological significance. Geoscience 27, 547 (in Chinese).
Cui, X., Zhu, W.-B., Ge, R.-F., 2014. Provenance and crustal evolution of the Northern Yangtze block revealed by detrital zircons from neoproterozoic-early paleozoic sedimentary rocks in the Yangtze Gorges area, South China. J. Geol. 122, 217–235.
Embleton, B.J.J., Schmidt, P.W., 1985. Age and significance of magnetizations in dolerite dykes from the Northampton block, western-Australia. Aust. J. Earth Sci. 32, 279–286.
Evans, D.A.D., Li, Z.X., Kirschvink, J.L., Wingate, M.T.D., 2000. A high-quality mid-Neoproterozoic paleomagnetic pole from South China, with implications for ice ages and the breakup configuration of Rodinia. Precambrian Res. 100, 313–334.
Fisher, R., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London Series A – Mathematical and Physical Sciences 217, 295–305.
Gao, W., Zhang, C., 2009. Zircon SHRIMP U-Pb ages of the Huangling granite and the tuff beds from Liantuo Formation in the Three Gorges area of Yangtze River, China and its geological significance. Geol. Bull. China 28, 45–50 (in Chinese).
Han, Z., Yang, Z., Tong, Y., Jing, X., 2015. New paleomagnetic results from Late Ordovician rocks of the Yangtze Block, South China, and their paleogeographic implications. J. Geophys. Res. Solid Earth 120, http://dx.doi.org/10.1002/2015JB012005
Harris, L.B., Li, Z.-X., 1995. Palaeomagnetic dating and tectonic significance of dolerite intrusions in the Albany Mobile Belt, Western Australia. Earth Planetary Sci. Lett. 131, 143–164.
Hofmann, M., Linnemann, U., Rai, V., Becker, S., Gärtner, A., Sagawe, A., 2011. The India and South China cratons at the margin of Rodinia—synchronous Neoproterozoic magmatism revealed by LA-ICP-MS zircon analyses. Lithos 123, 176–187.
Huang, K.N., Opdyke, N.D., Zhu, R.X., 2000. Further paleomagnetic results from the Silurian of the Yangtze Block and their implications. Earth Planet. Sci. Lett. 175, 191–202.
Hughes, N.C., Peng, S.C., Bhargava, O.N., Ahluwalia, A.D., Walia, S., Myrow, P.M., Parcha, S.K., 2005. Cambrian biostratigraphy of the Tal Group, Lesser Himalaya, India, and early Tsanglangpuan (late early Cambrian) trilobites from the Nigali Dhar syncline. Geol. Mag. 142, 57–80.
Jiang, B., Sinclair, H.D., Niu, Y., Yu, J., 2014. Late Neoproterozoic-Early Paleozoic evolution of the South China Block as a retroarc thrust wedge/foreland basin system. Int. J. Earth Sci. 103, 23–40.
Jiang, X., Wang, J., Cui, X., Zuo, K., Xiong, G., Lu, J., Liu, J., 2012. Zircon SHRIMP U-Pb geochronology of the Neoproterozoic Chengjiang Formation in central Yunnan Province (SW China) and its geological significance. Sci. China Earth Sci. 55, 1496–1507 (in Chinese).
Kendall, B., Creaser, R.A., Selby, D., 2006. Re-Os geochronology of postglacial black shales in Australia: constraints on the timing of "Sturtian glaciation. Geology 34, 729–732.
Kent, D.V., Zeng, X.S., Zhang, W.Y., Opdyke, N.D., 1987. Widespread late mesozoic to recent remagnetization of paleozoic and lower triassic sedimentary-rocks from south china. Tectonophysics 139, 133–143.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.
Lan, Z., Li, X.-H., Zhu, M., Zhang, Q., Li, Q.-L., 2015. Revisiting the Liantuo Formation in Yangtze Block, South China: SIMS U–Pb zircon age constraints and regional and global significance. Precambrian Res. 263, 123–141.
Li, Z.X., Evans, D.A.D., 2011. Late Neoproterozoic 40° intraplate rotation within Australia allows for a tighter-fitting and longer-lasting Rodinia. Geology 39, 39–42.
Li, Z.X., Zhang, L.H., Powell, C.M., 1995. South china in Rodinia – part of the missing link between Australia east Antarctica and Laurentia. Geology 23, 407–410.
Li, Z.X., Evans, D.A.D., Zhang, S., 2004. A 90 degrees spin on Rodinia: possible causal links between the Neoproterozoic supercontinent, superplurne, true polar wander and low-latitude glaciation. Earth Planet. Sci. Lett. 220, 409–421.
Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Res. 160, 179–210.
Li, Z.-X., Evans, D.A.D., Halverson, G.P., 2013. Neoproterozoic glaciations in a revised global palaeogeography from the breakup of Rodinia to the assembly of Gondwanaland. Sediment. Geol. 294, 219–232.
Lin, J.L., Fuller, M., Zhang, W.Y., 1985a. Paleogeography of the North and South China Blocks during the Cambrian. J. Geodyn. 2, 91–114.
Lin, J.L., Fuller, M., Zhang, W.Y., 1985b. Preliminary phanerozoic polar wander paths for the North and South China blocks. Nature 313, 444–449.
Liu, X., Gao, S., Diwu, C., Ling, W., 2008. Precambrian crustal growth of Yangtze craton as revealed by detrital zircon studies. Am. J. Sci. 308, 421–468.
Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.
Ma, G., Lee, H., Zhang, Z., 1984. An investigation of the age limits of the Sinian System in South China. Bull. Yichang Inst. Geol. Miner. Res., Chinese Acad. Geol. Sci., 1–29.
Macouin, M., Besse, J., Ader, M., Gilder, S., Yang, Z., Sun, Z., Agrinier, P., 2004. Combined paleomagnetic and isotopic data from the Doushantuo carbonates, South China: implications for the snowball Earth hypothesis. Earth Planet. Sci. Lett. 224, 387–398.
McElhinny, M.W., Powell, C.M., Pisarevsky, S.A., 2003. Paleozoic terranes of eastern Australia and the drift history of Gondwana. Tectonophysics 362, 41–65.
McFadden, P.L., 1990. A new fold test for palaeomagnetic studies. Geophys. J. Int. 103, 163–169.
McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from fisher distributions. Geophys. J. R. Astron. Soc. 67, 19–33.
McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism. Earth Planet. Sci. Lett. 87, 161–172.
McKenzie, N.R., Hughes, N.C., Myrow, P.M., Choi, D.K., Park, T.-y., 2011. Trilobites and zircons link north China with the eastern Himalaya during the Cambrian. Geology 39, 591–594.
Metcalfe, I., 2006. Paleozoic and Mesozoic tectonic evolution and palaeogeography of East Asian crustal fragments: the Korean Peninsula in context. Gondwana Res. 9, 24–46.
Opdyke, N.D., Huang, K., Xu, G., Zhang, W.Y., Kent, D.V., 1987. Paleomagnetic results from the Silurian of the Yangtze paraplatform. Tectonophysics 139, 123–132.
Pares, J.M., van der Pluijm, B.A., Dinares-Turell, J., 1999. Evolution of magnetic fabrics during incipient deformation of mudrocks (Pyrenees, northern Spain). Tectonophysics 307, 1–14.
Pisarevsky, S.A., Wingate, M.T.D., Stevens, M.K., Haines, P.W., 2007. Palaeomagnetic results from the Lancer 1 stratigraphic drillhole, Officer Basin, Western Australia, and implications for Rodinia reconstructions. Aust. J. Earth Sci. 54, 561–572.
Powell, C.M., Pisarevsky, S.A., 2002. Late Neoproterozoic assembly of East Gondwana. Geology 30, 4.
Schmidt, P.W., 2014. A review of Precambrian palaeomagnetism of Australia: palaeogeography, supercontinents, glaciations and true polar wander. Gondwana Res. 25, 1164–1185.
Schmidt, P.W., Williams, G.E., 1996. Palaeomagnetism of the ejecta-bearing Bunyeroo Formation, late Neoproterozoic, Adelaide fold belt, and the age of the Acraman impact. Earth Planet. Sci. Lett. 144, 347–357.
Schmidt, P.W., Williams, G.E., 2010. Ediacaran palaeomagnetism and apparent polar wander path for Australia: no large true polar wander. Geophys. J. Int. 182, 711–726.
Schmidt, P.W., Williams, G.E., 2013. Anisotropy of thermoremanent magnetisation of Cryogenian glaciogenic and Ediacaran red beds, South Australia: Neoproterozoic apparent or true polar wander? Glob. Planet. Change 110, 289–301.
Schmidt, P.W., Williams, G.E., McWilliams, M.O., 2009. Palaeomagnetism and magnetic anisotropy of late Neoproterozoic strata, South Australia: implications for the palaeolatitude of late Cryogenian glaciation, cap carbonate and the Ediacaran System. Precambrian Res. 174, 35–52.
Shen, Z.D.F., Wang, P., Tan, X., Wang, Z., 1999. Paleomagnetism of Early Triassic Daye formation and its tectonic implication. Chin. Sci. Bull. 44, 412–418.
Sohl, L.E., Christie-Blick, N., Kent, D.V., 1999. Paleomagnetic polarity reversals in Marinoan (ca, 600 Ma) glacial deposits of Australia: implications for the duration of low-latitude glaciation in Neoproterozoic time. Geol. Soc. Am. Bull. 111, 1120–1139.
Van der voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.
Wingate, M.T.D., Giddings, J.W., 2000. Age and palaeomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia-Laurentia connection at 755 Ma. Precambrian Res. 100, 335–357.
Xu, Y., Cawood, P.A., Du, Y., Hu, L., Yu, W., Zhu, Y., Li, W., 2013. Linking south China to northern Australia and India on the margin of Gondwana: constraints from detrital zircon U-Pb and Hf isotopes in Cambrian strata. Tectonics 32, 1547–1558.
Yang, Z.Y., Besse, J., 2001. New Mesozoic apparent polar wander path for south China: tectonic consequences. Journal of Geophysical Research-Solid Earth 106, 8493–8520.
Yang, Z., Sun, Z., Yang, T., Pei, J., 2004. A long connection (750–380 Ma) between South China and Australia: paleomagnetic constraints. Earth Planet. Sci. Lett. 220, 423–434.
Zhang, Q., Piper, J.D.A., 1997. Palaeomagnetic study of Neoproterozoic glacial rocks of the Yangzi block: palaeolatitude and configuration of south China in the late Proterozoic supercontinent. Precambrian Res. 85, 173–199.
Zhang, H., Zhang, W., Li, P., 1983. Paleomagnetism of the Sinian System of eastern Yangzi Gorges in China. Bull. Tianjin Inst. Geol. Min. Res. 6, 57–67 (in Chinese).
Zhang, S.H., Jiang, G.Q., Zhang, J.M., Song, B., Kennedy, M.J., Christie-Blick, N., 2005. U-Pb sensitive high-resolution ion microprobe ages from the Doushantuo Formation in south China: constraints on late Neoproterozoic glaciations. Geology 33, 473–476.
Zhang, S.B., Zheng, Y.F., Wu, Y.B., Zhao, Z.F., Gao, S., Wu, F.Y., 2006. Zircon U-Pb age and Hf isotope evidence for 3.8 Ga crustal remnant and episodic reworking of Archean crust in South China. Earth Planet. Sci. Lett. 252, 56–71.
Zhang, S., Jiang, G., Han, Y., 2008. The age of the Nantuo Formation and Nantuo glaciation in South China. Terra Nova 20, 289–294.
Zhang, S., Evans, D.A.D., Li, H., Wu, H., Jiang, G., Dong, J., Zhao, Q., Raub, T.D., Yang, T., 2013. Paleomagnetism of the late Cryogenian Nantuo Formation and paleogeographic implications for the South China block. J. Asian Earth Sci. 72, 164–177.
Zhang, S., Li, H., Jiang, G., Evans, D.A.D., Dong, J., Wu, H., Yang, T., Liu, P., Xiao, Q., 2015. New paleomagnetic results from the Ediacaran Doushantuo Formation in South China and their paleogeographic implications. Precambrian Res. 259, 130–142.
Zhu, B., Becker, H., Jiang, S.-Y., Pi, D.-H., Fischer-Goedde, M., Yang, J.-H., 2013. Re-Os geochronology of black shales from the Neoproterozoic Doushantuo Formation, Yangtze platform, South China. Precambrian Res. 225, 67–76.